

ATTACHMENT F

DATA REQUIREMENTS DELIVERABLES (DRDs)

03/19/2026

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Attachment F: Data Requirements
Deliverables (DRDs)
MTN-MGMT-CDRL-0001, Revision –
Mars Telecommunications Network

Mars Telecommunications Network (MTN) Data Requirements Deliverables (DRDs)



**Goddard Space Flight Center
Greenbelt, Maryland**

[illegible]

Table of TBDs/TBRs/TBSs

Action Item No.	Location	Summary	Individual/ Organization Actionee

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1 INTRODUCTION

Consistent with law, the National Aeronautics and Space Administration (NASA) is working toward a high-performance Mars telecommunications orbiter(s), capable of providing assets in and around Mars with robust, continuous communications with Earth.

1.1 Purpose

The purpose of this document is to define the data requirements that shall be delivered to the Government for the contracted work performed on the Mars Telecommunications Network (MTN) Project. The document provides descriptions, submission timeline, contents, and formats.

Throughout the project lifecycle, the MTN Project will gather information via technical insight and the data deliverables described herein are one aspect of how NASA MTN will obtain that information. These data deliverables are necessary to ultimately accept hardware fabricated for this project and make risk informed decisions throughout.

1.2 Scope

Subject to the Rights in Data clause, 52.227-14 Rights in Data-General (MAY 2014), this Data Procurement Document (DPD) sets forth the data requirements in each Data Requirements Description (DRD) and shall govern that data required by the DPD for the contract. The Contractor shall furnish data defined by the DRDs listed on the Contract Data Requirements List (CDRL) by category of data, attached hereto, and made a part of this DPD. Such data shall be prepared, maintained, and delivered to NASA in accordance with the requirements set forth within this DPD. In cases where data requirements are covered by a Federal Acquisition Regulation (FAR) or NASA FAR Supplement (NFS) clause, that clause shall take precedence over the DPD, consistent with clause FAR 52.215-8 Order of Precedence—Uniform Contract Format (OCT 1997).

1.3 Applicability

This data procurement document is applicable to the contracted work performed on the MTN Project.

1.4 Data Types for Contractual Efforts

The types of data and their contractual applicable requirements for approval and delivery are:

Type	Description
1	NASA reserves a time-limited right to disapprove in writing any versions and interim changes to those versions. The data shall be delivered to NASA for review not less than 20 working days prior to its release for use or implementation. The Contractor shall clearly identify the release target date in the “submitted for review” transmittal. If the Contractor has not been notified of any disapproval prior to the release target date, the data shall be considered approved. To be an acceptable delivery, disapproved data shall be revised to remove causes for the disapproval before its release.
2	NASA reserves a time-limited right to disapprove in writing any versions and interim changes to those versions. The data shall be delivered to NASA for review not less than 10 working days prior to its release for use or implementation. The Contractor shall clearly identify the release target date in the “submitted for review” transmittal. If the Contractor has not been notified of any disapproval prior to the release target date, the data shall be considered approved. To be an acceptable delivery, disapproved data shall be revised to remove causes for the disapproval before its release.
3	These data shall be delivered by the Contractor as required by the contract and do not require NASA approval. NASA may submit comments or questions on these documents for clarification or assistance with understanding of the DRD for the Contractor to consider. However, to be a satisfactory delivery, the data shall satisfy all applicable contractual requirements.

Table 1 Data Types Description

2 REFERENCE AND APPLICABLE DOCUMENTS

2.1 Applicable Documents

Applicable documents are those documents that form a part of this document. These documents carry the same weight as if they were stated within the body of this document.

Document Number	Document Title	Revision
NASA STD-871927	Implementing Planetary Protection Requirements for Spaceflight	

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2.2 Reference Documents

Reference documents are those documents that, though not a part of this document, serve to clarify the intent and contents of this document.

Document Number	Document Title	Revision
NFS 1852.204-76	Security Requirements for Unclassified Information Technology Resources	
NASA-STD-6016	Standard Materials and Processes Requirements for Spacecraft	C
NASA-STD-8739.14	NASA Fastener Procurement, Receiving Inspection, and Storage Practices for NASA Mission Hardware	
NASA-STD-8739.8	Software Assurance and Software Safety Standard	
NASA-STD 8719.24	NASA Payload Safety Requirements	A
MSFC-STD-3029	Guidelines for the Selection of Metallic Materials for Stress Cracking Resistance in Sodium Chloride Environments	A
MTN-402-REQ-0001	Mars Telecommunications Network (MTN) Objectives and Requirements Document	
NASA STD-871927	Implementing Planetary Protection Requirements for Spaceflight	
NPR 7120.5	NASA Space Flight Program and Project Management Requirements	F
NPR 7123.1	NASA System Engineering Processes and Requirements	D

3 LEVERAGING OF CONTRACTOR BEST PRACTICE/INTERNAL DATA DELIVERABLES.

Unless otherwise noted in this or other superseding documents, data products and documentation produced by the offeror through existing internal practices may be used to satisfy any number of data requirements described herein (to the extent practicable) granted that:

1. The supplier clearly communicates traceability to the data deliverables being satisfied by each submission and,
2. The NASA MTN reviewer concurs that the submission adequately satisfies the intent of required deliverable in question.

The DRD will call out any special format requirements if necessary. General format requirements for electronic and hardcopy are as follows. Note that the “pages” field of each DRD entry indicates how many pages are in each DRD description of this DPD; they do not indicate pages associated with submission.

3.1 Electronic Format

DRDs shall be delivered in their native format unless otherwise specified by the DRD or the Contracting Officer. A preferred format for each delivery item may be specified in the DRD, but the Contractor may utilize an alternative at the discretion of the CO.

Structured data delivered in satisfaction of a DRD is preferred to be in an open-source and standards-compliant format; proprietary file formats are acceptable, but each proprietary format submission shall include:

1. A definition of the application (including version) that produced the file.
2. The native proprietary file itself.
3. A commonly readable representation of the file (e.g. PNG or PDF).

If native file formats are not readable by COTS software and at the discretion of the CO, the Contractor may deliver the application (source code and executable) required in order to read the native format.

In general, files provided in an electronic format shall meet five basic requirements: “human readable”, “machine readable”, “text searchable”, “printable”, and “downloadable” utilizing publicly-available COTS software.

3.2 Hardcopy Format

Hardcopy submission of DRDs is necessary only at the discretion of the CO; in general, the Government shall accept electronic delivery of all data products.

4 DATA REQUIREMENTS LIST

Unless otherwise noted, days denote calendar days P-Preliminary B-Baseline F-Final ATP-Authority to Proceed							
DRD Number	DR Title	Type	Preliminary Design Phase	Detailed Design Phase	Fabrication, Build, Assembly & Test Phase	Launch Vehicle Integration & Post-Launch Phase	Frequency
DRD-PM-01	Project Management Plan	2	P-60 Days After ATP B-End of Phase				Update Upon Revision
DRD-PM-02	Integrated Master Schedule (IMS)	3	P-30 Days After ATP B-90 Days After ATP				Update Monthly
DRD-PM-03	Management Review Packages	3					Monthly
DRD-PM-04	Configuration and Data Management Plan	3	P-30 Days After ATP B-End of Phase				Update Upon Revision
DRD-PM-05	Risk Management Plan and Risk Management Reports	3	P-60 Days After ATP B-End of Phase				Risk reports Monthly
DRD-PM-06	Major Review Plans and Documentation	3					Plan-30 days before review Documentation-10 working days before review
DRD-PM-07	Offeror's IT Security Management Plan	2	B-6 Months After ATP				Update Upon Revision
DRD-PM-08	Insight and Oversight Implementation Plan	2	B-60 Days After ATP				Update Upon Revision
DRD-MD-01	Mission Operations Plan/ConOps	1	P-60 Days After ATP B-End of Phase	U	U	U	Updated at Major Milestones
DRD-MD-02	Mission Design Document	2	B-End of Phase	U	U	U	Updated at Major Milestones
DRD-SE-01	System Requirements	1	B-End of Phase	U	U	U	Updated at Major Milestones
DRD-SE-02	System Specification	2	B-End of Phase	U	U	U	Updated at Major Milestones
DRD-SE-03	Interface Control Documents and Drawings (ICDs)	2	B-End of Phase	F-End of Phase			
DRD-SE-04	Hardware Deliverable List	3	B-60 Days After ATP F-End of Phase				
DRD-SE-05	Acceptance Data Package of Delivered Hardware	1					P-20 Working Days prior to Pre-Ship Review F-20 Working Days prior to Acceptance Review
DRD-SE-06	Environmental Requirements Document	2	B-End of Phase				Update Upon Revision
DRD-SE-07	Design Data Book	2	P-End of Phase	U	U		F-20 Working Days prior to Acceptance Review
DRD-SE-08	Spacecraft Performance and System Resource Budget Reports	3	B-End of Phase	U	U		F-20 Working Days prior to Acceptance Review
DRD-SE-09	Assembly, Integration and Test (AIT) Plan	2	B-End of Phase	U	U	U	Updated at Major Milestones
DRD-SE-10	Verification & Validation Plan	2	B-End of Phase	U	U	U	Updated at Major Milestones
DRD-SE-11	Test Plans and Procedures	3					B-With Test Readiness Review data Package F-Within 10 Working Days of Test Completion
DRD-SE-12	Test Reports	3					F-Within 20 Working Days of Test Completion
DRD-SE-13	Verification Closure Notices and Validation Closure Notices/Requirements Verification Traceability Matrix	2	P-End of Phase	U	U		P-5 days after event (Notices) F-30 days after event(Notices) F-20 Working Days prior to Acceptance Review (Matrix)
DRD-SE-14	Drawing and Document Trees	3	P-End of Phase	B-End of Phase			Update Upon Revision
DRD-SE-15	Engineering Drawings, Models, and Associated Lists	3	0.3	0.9	F-End of Phase		
DRD-SE-16	Design and Construction Standards Utilized	3	B-End of Phase		F-End of Phase		
DRD-SE-17	Structural Analysis Reports	3	B-End of Phase		F-End of Phase		
DRD-SE-18	Thermal Analysis Reports	3	B-End of Phase	F-End of Phase			

Unless otherwise noted, days denote calendar days							
P-Preliminary B-Baseline F-Final ATP-Authority to Proceed							
DRD Number	DR Title	Type	Preliminary Design Phase	Detailed Design Phase	Fabrication, Build, Assembly & Test Phase	Launch Vehicle Integration & Post-Launch Phase	Frequency
DRD-SE-19	Power Analysis Report and Battery Usage Assessment	3	B-End of Phase	F-End of Phase			
DRD-SE-20	Electromagnetic Interference (EMI) Electromagnetic Compatibility (EMC) Control Plan	3	B-End of Phase				F-Within 20 Working Days of Pre-Ship Review
DRD-SE-21	Worst Case Analysis	3		B-End of Phase			F-Within 20 Working Days of Pre-Ship Review
DRD-SW-01	Software Requirements Specification (SRS)	3	P-End of Phase				F-Within 20 Working Days of Pre-Ship Review
DRD-SW-02	Master Data Dictionary	3	P-End of Phase				F-Within 20 Working Days of Pre-Ship Review
DRD-SW-03	Software Management and Development Plan	3	P-End of Phase	F-End of Phase			
DRD-SW-04	Software Design Description/Version Description Document	3	P-End of Phase				F-Within 20 Working Days of Pre-Ship Review
DRD-SMA-01	Request for Deviation or Waiver	1					As Needed
DRD-SMA-02	Safety and Mission Assurance (S&MA) Plan	2	P-End of Phase	B-End of Phase			Update Upon Revision
DRD-SMA-03	System Safety Analysis Report	2	P-End of Phase	B-End of Phase			F-Within 60 Working Days of Pre-Ship Review
DRD-SMA-04	Failure Modes and Effects Analysis and Critical Items List / Single Point Failure	2	P-End of Phase	B-End of Phase			F-Within 20 Working Days of Pre-Ship Review
DRD-SMA-05	EEE Parts Management Plan	3	P-End of Phase	B-End of Phase			F-Within 20 Working Days of Pre-Ship Review
DRD-SMA-06	Parts Identification List	3	P-End of Phase				F-Within 20 Working Days of Pre-Ship Review
DRD-SMA-07	Parts Stress/De-Rating Analysis	3	P-End of Phase	F-End of Phase			
DRD-SMA-08	Radiation Tolerance Report	3	P-End of Phase	F-End of Phase			
DRD-SMA-09	Nonconformance Processing and Corrective Action Plan and Nonconformance Reports	2	P-End of Phase	F-End of Phase			Reports as Needed
DRD-SMA-10	Materials Identification and Usage List/Material Usage Agreement	3	P-End of Phase	B-End of Phase			F-Within 20 Working Days of Pre-Ship Review
DRD-SMA-11	Planetary Protection Plan	2	P-End of Phase	F-End of Phase			
DRD-SMA-12	Safety Requirements Compliance Checklist (SRCC)	2	P-End of Phase	F-End of Phase			
DRD-COM-01	Communication System Analysis Report (DWE)	2	P-End of Phase	F-End of Phase			
DRD-COM-02	Communication System Analysis Report (UHF Prox)	2	P-End of Phase	F-End of Phase			
DRD-COM-03	Communication System Analysis Report (Advanced Prox)	2	P-End of Phase	F-End of Phase			
DRD-COM-04	Spectrum Management and Compliance Information	2	P-End of Phase	B-End of Phase			F-Within 20 Working Days of Pre-Ship Review
DRD-FO-01	Operations Data Book/User Guide	2	P-End of Phase	U	U		F-20 Working Days prior to Acceptance Review
DRD-FO-02	Flight Operations Procedures and Procedures Data	2	P-End of Phase	U	U		F-20 Working Days prior to Acceptance Review
DRD-FO-03	Mission Operations Plans	2	P-End of Phase	U	U		F-20 Working Days prior to Acceptance Review

5 DATA REQUIREMENTS DESCRIPTIONS

5.1 Data Requirements Description DRD-PM-01 (Project Management Plan)

Program:	MTN	DRD No.:	DRD-PM-01
Data Type:	2	Date Revised:	Initial Release
Pages:	3		

Title: Project management Plan

Description/Use: To describe the overall scope of the project and the implementation approach, including the systems, processes and methods, to provide overall coordination of management activities and plans for accomplishing the Contractor Performance Work Statement. The Project Management Plan (PMP) also describes the structure and environment within which the contract and subcontract(s) operate.

Initial Submission: Preliminary 60 days after Authority to Proceed (ATP).

Submission Frequency: Baseline version no later than end of Preliminary Design Phase and update as required afterward.

Remarks:

Contents: The PMP shall provide a description of the Contractor's management concepts, practices, approaches, plans, and schedules necessary for accomplishing (managing and controlling) the tasks described in the contractor Performance Work Statement (PWS). The PMP shall present those management systems to be utilized to define and delegate task assignments and shall define the roles and organizational relationships of the Contractor, Subcontractors, and Government. The PMP shall cover all aspects of contract and subcontract management for the contract. The PMP shall provide information giving the Government insight into staffing, organizational structure, approaches and processes used to manage activities across the contract. The PMP shall describe the methods the contractor will employ to provide Government data accessibility and deliverables.

The PMP shall identify and describe Control Plans that will be used in executing the PWS tasks. Control plans can either be a part of the PMP or separate stand-alone documents referenced in the appropriate part of the PMP.

In addition to the information described above, the PMP shall include, at a minimum, the following items:

1. Descriptions of the management approaches used to accomplish and monitor contractor and subcontractor tasks.
2. Description of roles and responsibilities of the contractor's key personnel.

3. Contractor Systems Engineering processes and tools used to implement the technical effort and document the overall technical approach, maintain integrity between NASA and Contractor requirements to the Contractor verification plans, and to transition the products through the lifecycle (for both Contractor and subcontractor efforts).
4. Management approach to routinely review and address cost, schedule, technical progress, concerns, and issues.
5. Description of major technical and milestone reviews, their purpose, and performance reporting to NASA in preparation for these reviews.
6. Communication Plan, including but not limited to a description of the overall interaction policy and approach, key interaction milestones, flexibility and adaptability of interactions, the process for continuous improvement and incorporations of lessons learned, and determination and implementation of cause and corrective actions.
7. Approach to interfaces with the Government, other Contractors, and/or other entities that are necessary and pertinent to the accomplishment of contractual tasks, including such things as data, analyses, equipment, software deliverables, schedules, interfaces, and other technical/managerial interactions.
8. Approach to the establishment and management of subcontracts.
9. Description of the contractor's acquisition strategy, including make/buy decisions and rationale.
10. Description of facility usage in support of the contract.
11. Performance Assessment Plan including, at a minimum:
 - a. Definition of performance assessment metrics that will be used to communicate to NASA the progress of the Contractor's efforts against the project baseline plan.
 - b. Description of management systems and processes used to implement the plan including the developing, timely gathering, compiling, maintaining, auditing, analyzing, reporting, and providing management review of performance metrics.
 - c. Description of the Technical Performance Parameters (TPPs) selected for monitoring during the design and development.
12. Schedule performance reporting plans which include schedule reserves.
13. Discuss approach to data rights and intellectual property (IP). Identify any limited data rights or restricted computer software issues, especially in the context of what the various corporate team members will provide. Explain how licensing agreements will be negotiated with vendors and how that data/IP will be treated, specifically identifying any requested special license agreements.
14. Define the Contractor Work Breakdown Structure (CWBS) which is a logical, hierarchical display of the subdivision of all project work to be completed. The CWBS shall include the approved element title and element number. The Contractor will use the CWBS to internally track program schedule and technical performance. The CWBS shall also be used when reporting technical and schedule performance to NASA.
15. Provide a CWBS Dictionary which may be a standalone document or an appendix to this plan. The CWBS dictionary shall describe and document the work content of every CWBS element and relevant efforts associated with each element (e.g., design, development, and manufacturing). The CWBS dictionary shall be arranged in the same order as the contract CWBS. The CWBS dictionary shall include the following for each CWBS element:
 - a. CWBS element title.
 - b. CWBS element code.
 - c. CWBS element content description (including quantities, relevant associated work, and contract end items where applicable).

- d. CWBS Index.
- e. PWS paragraph number.
- f. Specification (number and title) associated with the CWBS element (if applicable).
- g. Contract line item associated with the CWBS element.
- h. Date, revision number, revision authorization and approved changes.
- i. Contract Identification Number.

NOTE: It is not intended that this plan duplicate other plans called for in the Data Requirements List. This plan shall summarize the overall project and reference or summarize other plans where appropriate and shall reference Contractor internal procedures where applicable.

Format: Contractor's format is acceptable.

End of Data Requirements Description DRD-PM-01

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5.2 Data Requirements Description DRD-PM-02 (Integrated Master Schedule)

Program:	MTN	DRD No.:	DRD-PM-02
Data Type:	3	Date Revised:	Initial Release
Pages:	2		

Title: Integrated Master Schedule (IMS)

Description/Use: The IMS is the primary means of communicating project schedule information between the prime Contractor and the Government. The IMS shall use established standard processes, data structures and reporting conventions to plan, manage, and report the work required in the performance of this contract.

Initial Submission: Preliminary 30 days after Authority to Proceed (ATP).

Submission Frequency: Monthly within 5 working days after the close of the Contractor's accounting month. IMS shall be baselined no later than 90 days after ATP.

Remarks:

Contents: The IMS shall include all contractor milestones, activities, and tasks leading to and including all phases through MTN End of Mission (EOM). The IMS shall track and show percentage completion of all tasks. The IMS shall provide the ability to fully identify, analyze, mitigate and control scheduling risks and impacts; accurately identify and analyze critical path activities; clearly show schedule float (margin); and allow its users to easily measure the progress towards achieving MTN milestones.

Tasks and milestones from subcontractors shall be included with sufficient detail to develop a realistic critical path. Their data shall be consistent, statused monthly, and based on the same cutoff date to enable a realistic critical path.

Along with the IMS monthly deliverables, the Contractor shall provide a narrative description of any significant schedule variances or slips since the last month and the root cause of those slips and options for recovery of that time in the remaining work left on the schedule. The narrative shall also track the status of previously reported variances or slips and the progress made to resolve them. Schedule variances or slips that impact the critical path shall be part of this narrative. Any other variances or slips that are of concern to the Contractor and/or have to potential to impact the critical path in the future should also be reported at the discretion of the Contractor.

Relationships/Dependencies: All discrete tasks/milestones except for the start and end of the contract or for interim receipt and delivery events to/from external entities shall have at least one predecessor and successor.

Schedule Margin: If schedule margin is included in the IMS, it must be as a defined task and clearly labeled as “Schedule Margin” or contingency. The number of schedule margin tasks should be minimal with most of the margin duration placed as the last task before the ending contract event. Other acceptable locations for placing smaller amounts of schedule margin are prior to high level project milestones.

Additionally, at a minimum, the IMS shall be formatted with the largest increment of time being measured in units of months.

Summary Master Schedule: The schedule shall include a top level Gantt chart summary arranged by WBS and that reflects all contract and controlled milestones, major program/project phases (i.e. design, fabrication, integration, assembly, test, etc.) and all end-item deliveries. It shall reflect, either by manual creation or by automated summarization, a vertically integrated rollup of intermediate and detailed schedule data.

Format: IMS shall be provided in Microsoft Project file format.

End of Data Requirements Description DRD-PM-02

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5.3 Data Requirements Description DRD-PM-03 (Management Review Packages)

Program:	MTN	DRD No.:	DRD-PM-03
Data Type:	3	Date Revised:	Initial Release
Pages:	2		

Title: Management Review Packages

Description/Use: To establish activity reporting for the Contractor's progress and accomplishments. The Management Review packages support the monthly management reviews of schedule and technical performance between the Contractor and NASA.

Initial Submission: 12 days after the close of the contract month that is subsequent to the month in which the post-award kick-off occurs.

Submission Frequency: Monthly within 5 business days after the close of the Contractor's accounting month.

Remarks: Management reviews shall be held monthly, face to face management reviews should be held bi-monthly at the Contractor's facility,

Contents: The Management Review data packages support monthly reviews of schedule and technical performance. Metrics that effectively indicate the level of success in the execution of contract requirements and the status of the Contractor's achievements shall be presented at the review. The Technical Performance Parameters shall be reported by the Contractor to status their margin growth/loss. All metrics shall be developed and defined by the Contractor with concurrence from the Government. The format provides a standardized approach for review materials.

The monthly Review package shall include, at a minimum:

1. Summary Section
 - a. Pertinent Performance/Outcome Based metrics
 - b. Technical Performance Parameters (including margin growth/loss and forecast)
 - c. Stoplight Status of schedule, and technical performance.
 - d. Status of risks and opportunities.
 - e. Cumulative variance explanations (to fiscal year plan) and End-Of- Year trend variance explanations (if threshold criteria exceeded).
 - f. Performance Work Statement Reconciliation (i.e., Special Projects, etc. that need to be accounted for) (as required)
 - g. Status of DRDs that are due in the next 90 days
 - h. Areas of Emphasis Summary
 - i. Major Contract Milestone Schedule - 90 Day
2. Component Sections (done at major subsystem level):

- a. Include all Summary Section items.
 - b. Accomplishments
 - c. Pertinent Next Level Performance/Outcome Based metrics (include any other Yellow or Red metrics)
 - d. Details of Technical and schedule issues, risks and opportunities
 - e. Status of document and drawing release/approval as well as engineering change reports.
3. Safety and Mission Assurance General
S&MA Status:
- a. Status of significant activities relating to the accomplishments of the PWS S&MA tasks. Risks to the accomplishment of PWS tasks shall also be identified.
 - b. S&MA key issues and concerns, including impact, action plan (including any NASA help needed), and projected resolution date.
- Detailed content related to S&MA disciplines reliability, EEE parts, quality assurance, materials and processes, system safety, and software assurance:
- a. Status of analyses with traceability to assembly/subsystem/system
 - b. Summary of all activities for that period to include issues/resolutions, analyses results, test results, nonconformances, failures, anomalies, waiver/deviations, audits & findings, mishaps, and quality metrics (including software)
 - c. Summary of upcoming activities for the next period
 - d. Status of corrective actions and open items

Format: Contractor's format is acceptable

End of Data Requirements Description DRD-PM-03

Released under the
NASA Open Source
Policy

5.4 Data Requirements Description DRD-PM-04 (Configuration and Data Management Plan)

Program:	MTN	DRD No.:	DRD-PM-04
Data Type:	3	Date Revised:	Initial Release
Pages:	2		

Title: Configuration and Data Management Plan

Description/Use: To describe the Contractor's method for accomplishing the configuration and data management requirements of the contract.

Initial Submission: Preliminary 30 days after Authority to Proceed (ATP).

Submission Frequency: Baseline version no later than end of Preliminary Design Phase and update as required afterward.

Remarks: This DRD may be satisfied by inclusion of required content in DRD-PM-01 Project Management Plan submissions. There is no requirement for a separate, stand-alone document.

Contents: NOTE: For this DRD, a Contractor's standard document can be submitted if it meets the intent of the content requirements of this deliverable.

The CDMP shall provide a section that defines the Contractor's data management approach. The data management section shall include, at a minimum:

1. Relationship of the Contractor's management, organization and planning to the data management program and the Contractor's other administrative and technical organizations.
2. Configuration Management Planning; Configuration Identification, which includes Interface Management; Configuration Change Control; Configuration Status Accounting, which includes Traceability; Configuration Verification and Audits; and Technical Data Management.
 - a. Configuration status accounting system shall provide information defining the as-designed and as-built configuration of the system and the status of changes to this configuration, provide traceability of changes to controlled products, serve as the basis for communicating accurate and timely status of all Configuration Items, and ensure the delivered documentation accurately describes and represents the associated Configuration Items.
3. Integration of the CM and DM systems across all pertinent project elements such as modeling and simulation, manufacturing, operations, quality, as well as acquisition organizations including customer and vendor Configuration and Data Management (CDM) systems.
4. Specific management policies, practices and work instructions to be used in implementing the data management program include the following elements:
 - a. Control procedures including management of MTN project directives and change control
 - b. Storage and retrieval procedures
 - c. Subcontractor control procedures
 - d. Special restrictions

- e. Government access to data generated during contract performance using an interactive collaborative environment (or equivalent)
 - f. Preliminary and final data submittal schedules for fulfilling submittal of DRD data in the specified quantities, specific media (electronic, paper, other), and due dates required.
5. Records retention approach.

Format: Contractor's format is acceptable

End of Data Requirements Description DRD-PM-04

Released Version

5.5 Data Requirements Description DRD-PM-05 (Risk Management Plan and Risk Management Reports)

Program:	MTN	DRD No.:	DRD-PM-05
Data Type:	3	Date Revised:	Initial Release
Pages:	2		

Title: Risk Management Plan and Risk Management Reports

Description/Use: Provides a baseline document for planning, management, control, and implementation of the Contractor's risk management.

Initial Submission: Preliminary 60 days after Authority to Proceed (ATP).

Submission Frequency: Risk Management Plan baseline version no later than end of Preliminary Design Phase and update as required afterward. Risk Reports submitted monthly within 5 business days after the close of the Contractor's accounting month.

Remarks: Risk Management Plan may be satisfied by inclusion of required content in DRD-PM-01 Project Management Plan submissions. There is no requirement for a separate, stand-alone document.

Contents: NOTE: For the Risk Management Plan portion of this DRD, a Contractor's standard document can be submitted as long as it meets the intent of the content requirements of this deliverable.

At a minimum, the Risk Management Plan (RMP) shall include:

1. **Introduction:** specify the program/project risk objectives and policy toward risk. Explain the purpose, scope, assumptions, constraints, key ground rules, and policy pertaining to the project risk management process.
2. **Overview of Process:** provide an overview of the risk management process and information flow; describe how the risk management process integrates and relates to other project management and system engineering activities. Include general risk mitigation strategies to be employed throughout project life cycle.
3. **Organization:** show the organization, roles, and responsibilities of program, project, customer, and supplier key personnel with regard to risk management. Document how team members will be trained in the application of risk management methodology.
4. **Process Details:** provide the risk management process details and related procedures, methods, tools, and metrics. Include here, or in an appendix, the specific methodologies to be used for activities of risk management: identify, analyze, plan, track, control, communicate and document as well as de-scope processes and risk acceptance versus closure criteria. Include the process to be used for continual assessment of the project risk profile. Describe how risk information will be communicated both internally to the project staff and throughout the NASA management chain.
5. **Documentation of Risks:** specify the format and data elements that will comprise the project risk list (and/or risk database), how configuration control will be applied, and how the list will be used and updated. Explain how team members will be able to access the current risk list at any time.

6. Key System Interfaces: describe the relationship and interfaces of the risk management process to lessons learned reporting, System Safety analyses, reliability/probabilistic risk analyses, schedule, financial and other business reporting systems.
7. Appendix: material that is too detailed or sensitive to be placed in the main body of text may be placed in an appendix or included as reference. Include the appropriate reference in the main body of the text. Appendices may be bound separately, but are considered to be part of the RMP document. Include an alphabetized list of the definitions for abbreviations and acronyms used in this document. Include an alphabetized list of definitions for special terms used in the document, i.e., terms used in a sense that differs from or is more specific than the common usage for such terms. Include a risk attributes scorecard that provides the criteria for establishing likelihood and consequence values of a risk.

At a minimum, the Risk Report shall include:

1. Risk Focus Charts which include:
 - a. Risk identification number
 - b. Risk statement
 - c. Risk context
 - d. Initial likelihood of occurrence score and rationales consistent with the risk attribute scorecard values established in the RMP
 - e. Initial consequence of occurrence score with safety, schedule, and technical/performance rationales consistent with the risk attribute scorecard values established in the RMP
 - f. Risk status
 - g. Timeframe to begin mitigations
 - h. Risk approach including mitigation plans (if applicable) and metrics with resulting likelihood and consequence scores consistent with the risk attribute scorecard values established in the RMP
2. Summary of risk mitigation activity and progress including objective evidence of exit criteria accomplishments
3. Applicable schedule for mitigation activity
4. Documentation of contingency planning for risks that could not be successfully mitigated, accepted, or closed
5. Risk attribute scorecard as established in the RMP
6. List of any closed or accepted risks since the last report and rationale for closure or acceptance
7. Consequence based 5x5 risk matrix
8. Risk list including risk trending, risk identification number, approach, and risk title

Format: Contractor's format is acceptable

End of Data Requirements Description DRD-PM-05

5.6 Data Requirements Description DRD-PM-06 (Major Review Plans and Documentation)

Program:	MTN	DRD No.:	DRD-PM-06
Data Type:	3	Date Revised:	Initial Release
Pages:	1		

Title: Major Review Plans and Documentation

Description/Use: Provides a baseline document for planning, management, control, and implementation of the Contractor's risk management.

Initial Submission: See Submission Frequency

Submission Frequency: Plans to be provided 30 days prior to review. Documentation to be provided 10 working days before review.

Remarks: Some portions of this deliverable may be satisfied by inclusion of required content in DRD-PM-01 Project Management Plan submissions.

Contents: NOTE: For this DRD, a Contractor's standard document can be submitted as long as it meets the intent of the content requirements of this deliverable.

For each review, the Contractor shall prepare a Review Plan that defines roles and responsibilities, entrance and exit success criteria, what data will be delivered in support of each review (including DRDs), and how comments will be collected, dispositioned and distributed. The review plans should also address verification, qualification, and validation of heritage hardware and software for its intended use on MTN.

For each review, the Contractor shall prepare and submit a review package, presentation material, and the supporting data necessary for the complete understanding and assessment of the review in accordance with the approved Review Plan. In addition to the entrance and success criteria and requirements, required DRDs will be reviewed

Format: Contractor's format is acceptable

End of Data Requirements Description DRD-PM-06

5.7 Data Requirements Description DRD-PM-07 (Information Technology Security Management Plan)

Program:	MTN	DRD No.:	DRD-PM-07
Data Type:	2	Date Revised:	Initial Release
Pages:	3		

Title: Information Technology Security Management Plan

Description/Use: To define, document, and implement the security requirements of the Contractor's Information Technology (IT) system as well as to delineate the responsibilities and expected behavior of all individuals who access the system. The Information Technology Security Management Plan shall be viewed as documentation of the structured process of planning adequate, cost-effective security protection for the system. This shall ensure the proper security controls are architected into the overall end product for easier potential transition to NASA.

Initial Submission: Within 6 months after Authority to Proceed (ATP).

Submission Frequency: Update as Required

Remarks:

Contents: NOTE: For this DRD, a Contractor's standard document can be submitted as long as it meets the intent of the content requirements of this deliverable.

The Contractor shall implement and maintain a National Institute of Standards & Technology (NIST) Risk Management Framework-Based IT System Security Plan for all systems provided or operated in performance of this contract as required in clause I.12, NFS 1852.204-76, Security Requirements for Unclassified Information Technology Resources.

The IT System Security Plan provides an overview of the security requirements of the system and describes the implementation details of the management, operational, and technical security controls in place or planned, and responsibilities and expected behavior of all individuals who access the system.

The IT System Security Plan, at a minimum, shall include the following:

A description of the detailed implementation of the following classes of control:

1. Operational Controls - The security controls (i.e., safeguards or countermeasures) for an information system that are primarily implemented and executed by people (as opposed to systems).
 - a. Documentation of operational controls may address the following areas, as applicable to the system, including implementation approach, responsible personnel or roles, execution schedules or frequencies, documentation maintained, and methods for validating effectiveness:

- i. Contingency Planning - Backup and recovery procedures, alternate processing arrangements, contingency plan testing
 - ii. Incident Response - Incident detection, classification, response procedures, reporting and escalation
 - iii. Maintenance - Scheduled maintenance procedures, maintenance personnel controls, maintenance logging
 - iv. Physical and Environmental Protection - Physical access controls, visitor management, environmental monitoring
 - v. Personnel Security - Position screening requirements, access authorization/termination procedures, security responsibilities
 - vi. Security Awareness and Training - Security awareness programs, role-based training requirements, training records
 - vii. Media Protection - Media handling, storage, transport, sanitization, and disposal procedures
- 2. Management Controls - Actions taken to manage the development, maintenance, and use of the system, including system-specific policies, procedures and rules of behavior, individual roles and responsibilities, individual accountability, and personnel security decisions.
 - a. Documentation of management controls may address the following areas, as applicable to the system, including implementation approach, responsible personnel or roles, execution schedules or frequencies, documentation maintained, and methods for validating effectiveness:
 - i. Planning - Security and privacy plans, system security plan development and updates, rules of behavior
 - ii. Risk Assessment - Risk assessment processes, vulnerability identification and analysis, risk determination and mitigation strategies
 - iii. System and Services Acquisition - System development life cycle management, security requirements in acquisition processes, developer security testing and evaluation
 - iv. Program Management - Information security program planning, critical infrastructure plans, security program oversight
 - v. Assessment, Authorization, and Monitoring - Security control assessments, authorization decision processes, continuous monitoring strategies
 - vi. Supply Chain Risk Management - Supply chain risk assessments, supplier security requirements, provenance tracking
- 3. Technical Controls - Security controls (i.e., safeguards or countermeasures) for an information system that are primarily implemented and executed by the information system through mechanisms contained in the hardware, software, or firmware components of the system. The controls can provide automated protection for unauthorized access or misuse, facilitate detection of security violations, and support security requirements for applications and data.
 - a. Documentation of technical controls may address the following areas, as applicable to the system, including implementation approach, responsible personnel or roles, execution

schedules or frequencies, documentation maintained, and methods for validating effectiveness:

- i. Access Control - Account management, access enforcement mechanisms, least privilege implementation, remote access controls
- ii. Identification and Authentication - User identification and authentication, device authentication, multi-factor authentication
- iii. Audit and Accountability - Audit logging, audit record content and generation, audit review and analysis, audit record retention
- iv. System and Communications Protection - Boundary protection, cryptographic protections, network segmentation, transmission confidentiality and integrity
- v. System and Information Integrity - Flaw remediation, malicious code protection, system monitoring, security alerts and advisories
- vi. Configuration Management - Baseline configurations, configuration change control, security impact analysis of changes

Format: Contractor's format is acceptable

End of Data Requirements Description DRD-PM-07

Released Version

5.8 Data Requirements Description DRD-PM-08 (Insight and Oversight Implementation Plan)

Program:	MTN	DRD No.:	DRD-PM-08
Data Type:	2	Date Revised:	Initial Release
Pages:	2		

Title: Insight and Oversight Implementation Plan

Description/Use: NASA will utilize the Contractor's Insight Implementation Plan to ensure NASA personnel, and its support services contractors can perform their insight and oversight.

Initial Submission: Baseline within 60 days after Authority to Proceed (ATP).

Submission Frequency: Update as Required

Remarks:

Contents: The Contractor's Insight and Oversight Implementation Plan shall provide a description of the Offeror's proposed organizational structure, interfaces, management approach, assignment of key personnel, major subcontractor and supplier teaming arrangements, and approach to communications and coordination with NASA. The Contractor's Insight and Oversight Implementation Plan shall also describe the accommodations for providing Government personnel and its support services contractor(s) timely and open access necessary to obtain a working-level understanding into all Contractor's activities leading to hardware delivery and ultimately sustained operations, including the approach to accommodating Government insight of technical and programmatic performance, including safety and mission assurance while still allowing the Contractor to maintain the efficiencies and effectiveness of their normal commercial processes. These accommodations include design review board membership, facility access, data, and any other information including technical and management processes required to meet contract requirements, milestone acceptance criteria, and to support approval of Type 1 and 2 data deliverables. The Contractor's Insight and Oversight Implementation Plan shall identify instructions or training required to allow for meaningful NASA insight and participation (e.g., facility and safety familiarization).

In addition, the Insight and Oversight Implementation Plan shall address:

1. Accommodating Government Insight and the following items:
 - a. The transparency by which the Contractor will provide the Government on-going access into activities and data used for mutually addressing risks and achieving successful milestone acceptance including membership on system level design review boards, approval of Type 1 and 2 data deliverables and making progress toward an endorsement of flight readiness and mission success
 - b. The transparency by which the Contractor will provide the Government on-going access to data, both remotely and on-site, in a useable and readable format; and provide the Government the ability to locate and review all data used in performance of this contract, to include technical data, supporting data/information, and administrative and management information with the exception of financial information. Whether the data accessed includes

- the ability of the Government to download or copy data; the Contractor's proposed data restriction legends on accessed data; and mechanisms to ensure that access data is not confused with data delivered under the contract.
- c. The timeframe the Contractor shall notify the NASA MTN designee of technical meetings, control boards, reviews, demonstrations and tests to permit meaningful Government participation through the entire event.
2. Accommodating Government Quality Assurance (GQA) functions. This shall include:
 - a. The timeliness and ease by which the Contractor will accommodate NASA and make data available for the Government to perform a successful risk based analysis that will facilitate the identification of high risk areas.
 - b. The timeliness and ease by which the Contractor will accommodate NASA performing the GQA activities identified by the risk based analysis and make data available to achieve successful GQA activities. GQA refers to the various functions performed by the Government Safety and Mission Assurance personnel to determine whether a contractor has fulfilled contract obligations pertaining to quality. The term quality, when used within the context of this DRD, refers to compliance with requirements that are levied upon the Contractor, including, but not limited to, safety, technical, item configuration, reliability, and quality management system requirements. GQA functions are planned and conducted on the basis of contract risk, to achieve confidence levels commensurate with the severity of consequences associated with noncompliance and to mitigate circumstances where there is an elevated likelihood of noncompliance. Audits or surveys are the typical methods used.
 3. Establishing a cooperative environment between NASA and the contractor dedicated to successful endorsement of flight readiness and mission success of MTN. This shall include:
 - a. Establishing excellent working relationships at every level of both organizations that enable both parties to solve problems as a team.
 - b. Ensuring a level of candor that permits the parties to mutually avoid risk and enables a continuous dialogue.
 4. Resolving concerns and issues identified by NASA and its support services contractor's insight effort. This shall include:
 - a. The process to permit timely elevation of issues to NASA.
 - b. The process of highlighting critical activities of interest (such as changes, decisions, key testing events, critical meetings) to NASA.
 5. Accommodating NASA's insight in the Contractor's development and successful fulfillment of operational requirements. This shall include:
 - a. The timeliness and ease by which the Contractor will accommodate NASA's insight in the operational activities and make data available to mutually address risks associated with achieving successful demonstrations, tests, completing acceptance milestone review criteria and a successful endorsement of flight readiness and mission success of MTN.
 - b. The process of highlighting critical activities of interest (such as changes, decisions, key events, critical meetings) to NASA.
 - c. The process to permit timely elevation of issues to NASA.
 6. In addition to access to Contractor facilities, the Contractor shall describe the NASA provisions for office space co-located on-site consistent with the proposed level of insight that would facilitate communication with NASA. This includes items such as, badging, furniture, telephones, and use of

easily accessible fax, data lines, and copy machines, for full-time and temporary Government insight and support services contractor personnel in performance of this contract, including training. This shall include:

- a. Co-located on site accommodations clearly identified.
 - b. Innovative use of technology that enables effective participation for completing milestone reviews acceptance criteria and making progress toward a successful endorsement of flight readiness and mission success of MTN.
7. The process of providing the Government insight into all subcontractors and suppliers performing or supporting any critical work associated with this contract.

Format: Contractor's format is acceptable

End of Data Requirements Description DRD-PM-08

Released Version

5.9 Data Requirements Description DRD-MD-01 (Mission Design Plan /Concept of Operations)

Program:	MTN	DRD No.:	DRD-MD-01
Data Type:	1	Date Revised:	Initial Release
Pages:	4		

Title: Mission Design Plan Plan/Concept of Operations

Description/Use: The MTN Concept of Operations (ConOps) defines the overall high-level concept of how the MTN will be used and operated. Timelines and sequencing should be defined for the set of operational cases or modes. The ConOps establishes a framework for the MTN requirements and defines the architecture of the ground and spaceflight operations addressing both the contractor's and NASA's objectives.

The Mission Design Plan will include the planning and approach of the MTN mission design team to the development and execution of building and delivering the reference, nominal trajectory as well as all off-nominal trajectory analysis to the MTN spacecraft. The trajectories to be modeled include the insertion trajectory into Martian orbit.

The Mission Design Plan will also include the description of the tools that will be used to develop and optimize the trajectory that the MTN will fly to the destination Martian orbit. The Mission Design Plan will describe the plan to model missed thrust and develop a mitigation strategy in trajectory design, and low thrust margin planning in both propellant and time. The plan will describe additional analysis such as Monte Carlo simulation and analysis, used in developing the nominal mission event timeline.

The objective of Mission Design Plan will be to obtain high-accuracy models and realistic simulations by detailing the approach to utilize a combination of analyses/modeling/simulation on-orbit performance and operating characteristics. These models and simulations can use testing and flight data obtained to assess, improve, and validate.

Initial Submission: Preliminary within 60 days after Authority to Proceed (ATP).

Submission Frequency: Baseline during Preliminary Design Phase, Updated at major milestones as identified in DRD-PM-02 (Integrated Master Schedule).

Remarks:

Contents:

The MTN Concept of Operations should include:

1. Contractor and NASA Spaceflight Demonstration Objectives
2. Major phases starting with launch and ending upon MTN end of mission.
3. Operation timelines

4. Operational phases or modes
5. Phase or mode transitions
6. Communications strategy
7. Critical Events
8. Identification of set of operations supporting spaceflight demonstration objectives
9. Driving off-nominal, contingency, or degraded-mode operational situations
10. Any other sections pertaining to the relay service ConOps

The Mission Design Plan document shall include at a minimum, the outline below:

1. Mission Design Team Description
 - a. Team makeup and description
 - b. Contractor/NASA Government mission design team interaction and relationship
 - c. Trajectory analysis expertise and skill mix
 - d. Mission design team operations
 - i. data sharing tools and formatting
 - ii. virtual and physical meeting cadence
2. Description of the tools used to perform the various aspects of the trajectory modeling.
 - a. Low thrust trajectory tools selection and description
 - b. Additional non-trajectory analysis Tools selection and description
 - c. Custom Tool development description
 - d. Data transfer description
 - i. between tools,
 - ii. between team members,
 - iii. between launch vehicle and MTN Spacecraft,
 - iv. between ground and spacecraft
 - e. Tool file formats and outputs
 - i. Trajectory output formats
 - ii. Trajectory development File formats
 - iii. Interoperability between tools
3. Trajectory Development Plan: Description of the plan to develop the different phases of the trajectory. How will the analysis be carried out? Description of the assumptions that will be applied to the phase of the mission. During the following phases of the mission, provide the following analysis plan and assumptions.
 - a. Launch Phase (covers from launch to MTN spacecraft separation)
 - i. Nominal trajectory development plan
 - ii. Launch window Analysis
 - iii. Missed Thrust plan
 - iv. Margin plan
 - v. Monte Carlo analysis
 - vi. Perigee raise assumptions
 - vii. Integrated mission optimization
 - viii. Development of mission specific launch requirements
 - b. Cruise to Mars and Orbit Capture
 - i. Insertion phase
 - ii. Nominal low thrust trajectory Analysis

- iii. Missed Thrust plan
- iv. Trajectory Margin plan
- v. Monte Carlo analysis
- vi. Ground station contact analysis
- c. Normal Routine Operations
 - i. Navigation errors
 - ii. Secondary requirements and assumptions
 - iii. Margin plan (propellant, time)
 - iv. Monte Carlo analysis
 - v. Ground station contact analysis
 - vi. Mission optimization
- 4. Trajectory Validation and Verification Plan
 - a. Requirements identification
 - b. Plan for requirements verification and validation
 - c. Plan for Trajectory verification and validation
 - d. Mission Design Panel description, makeup and purpose
 - e. Frequency of Mission Design Panels
 - f. Documents and assessments required for review
- 5. MTN system modeling and simulation assumptions necessary for trajectory modeling for all tools, models, simulations, and analyses that are used the following information, at a minimum, shall be delivered along with the system tools (as applicable):
 - a. All documented assumptions and abstractions underlying the conceptual model, including their rationales and/or reference documents.
 - b. Source code and build files for all models. Model in native format (m-files, Excel spreadsheet, .spk, .ideck, etc.). Software version number, including patch level, and compiler, if applicable.
 - c. Basic structure and mathematics of the model/tool (e.g. reality modeled, equations solved, behaviors modeled, conceptual models).
 - d. Required units and vector coordinate system/frames (where applicable) for all input/output variables.
 - e. Limits of operations of models/tools.
 - f. Methods of uncertainty quantification and the uncertainty in any data used to develop the model/tool or incorporated into the model/tool.
 - g. Guidance on proper use of the model/tool.
 - h. Any parameter calibrations and the domain of calibration.
 - i. Updates of models/tools (e.g. solution adjustment, change of parameters, calibration, and test cases) and assign unique version identifier, description, and the justification for the updates.
 - j. Obsolescence criteria and obsolescence date of the model/tools (obsolescence refers to situations where changes to the real system invalidate the model).
 - k. Example test cases
 - l. Compiled versions that run under one of Windows, Macintosh, Linux and UNIX variants.

- m. Source code for tools used to create lookup tables within the models, unless they were created with Commercial-off-the-shelf (COTS) software.
- 6. Inputs Mission Design Operations Plan
 - a. Description of the inputs needed to the mission design operations

Format: Documents to be provided as a native Word document, editable version as well as a pdf. Modeling and simulations shall include source code and build (also referred to as input) files for all models. Models provided in native format. Also to be included are summary excel (.xls) or .csv file of all tabular data used to create plots within the document(s) and associated native files of the reference trajectory and applicable orbits. Details such as the number and frequency of the time steps taken in generating the files are to be determined, and documented, and will be chosen to gather as much data as necessary to allow for independent reproduction of the trajectory data. Also provide compiled versions that run under one of Windows, Macintosh, Linux and UNIX variants and example test cases. Provide source code for tools used to create lookup tables within the models, unless they were created with COTS software.

End of Data Requirements Description DRD-MD-01

5.10 Data Requirements Description DRD-MD-02 (Mission Design Document)

Program:	MTN	DRD No.:	DRD-MD-02
Data Type:	2	Date Revised:	Initial Release
Pages:	2		

Title: Mission Design Document

Description/Use: The purpose of the Mission Design Document (MDD) is to provide detailed descriptions of a set of mission studies that evaluate the feasibility of the use of the MTN in the Mars orbit for Government both to maintain the final orbit and for mars missions occurring after the insertion into the final mars orbit, during the Mars Operations Phase in Martian space. This document will also provide an independent check to the NASA mission design team on the modeling of the various mission phases: Earth Orbit Raise (EOR) (if applicable), Orbit maintenance in Mars and representative Martian transfers from the nominal Mars orbit, for events like EDL. This relevant data include a description of maneuvers necessary to maintain the MTN in Mars orbit, and example mission concepts illustrating the use of the MTN to both transit from the Launch Vehicle insertion into the desired Mars orbit as well as examples of transfer between nominal Mars orbit and support for an EDL demo. The document will describe any potential excursion missions to and from a selected reference orbit. This document will provide a discussion of mission design considerations, performance information, and mission parameter sensitivities associated with these mission scenarios. The ultimate intent of this document is to provide an avenue to supply the government team with independent trajectory design that will be used to verify the government reference missions.

Initial Submission: Baseline by the end of Preliminary Design Phase

Submission Frequency: Updated at major milestones as identified in DRD-PM-02 (Integrated Master Schedule).

Remarks:

Contents:

Each delivery for this DRD shall consist of

1. A revision of the Mission Design Document consisting of the details described below.
2. An Excel-based manifest listing each model included in the delivery along with the model version and a brief description.

The contents of the Mission Design Document shall consist of the following details:

1. Introduction
2. Scenario overview
3. MTN Spacecraft assumptions

-
- a. Mass
 - b. Power system
 - c. Propulsion System
 - d. Vehicle summary
4. Insertion trajectory
- a. Reference orbit description
 - b. Launch Vehicle description and performance
 - c. Launch window analysis
 - d. Missed thrust and margin policy
 - e. Mission event timeline
 - f. Distance to Earth vs mission timeline
 - g. Distance to Mars vs mission timeline
 - h. Mass history
5. Orbital operations
- a. Mission outline
 - b. MTN orbit description
 - c. MTN mission option summary
 - d. Mission event timeline
 - e. Distance to Earth vs mission timeline
 - f. Distance to Mars vs mission timeline
 - g. Propellant budget summary
 - h. Mass history

Format: Native Word document, editable version and pdf.

End of Data Requirements Description DRD-MD-02

5.11 Data Requirements Description DRD-SE-01 (System Requirements)

Program:	MTN	DRD No.:	DRD-SE-01
Data Type:	1	Date Revised:	Initial Release
Pages:	2		

Title: System Requirements

Description/Use: The purpose of the MTN Flight System Requirements document is to define the system requirements and Verification Methods (A, D, I, T). The MTN Flight System Requirements and subsystem requirements establish a formal description of functional and performance requirements of the MTN flight hardware and software elements. These requirements will form the basis for assessing the adequacy of the MTN flight system design to meet the mission objectives.

Initial Submission: Baseline by the end of Preliminary Design Phase

Submission Frequency: Updated at major milestones as identified in DRD-PM-02 (Integrated Master Schedule).

Remarks:

Contents:

The MTN System Requirement set documents a complete set of MTN spacecraft system and subsystem level requirements that envelope all functional, performance, interoperability, and Safety & Mission Assurance-driven requirements for the MTN launch and ascent, spaceflight and operational phase. The requirements set shall include the following attributes:

1. Each requirement shall have a unique identifying requirement number,
2. Each requirement shall have a unique title,
3. Each requirement shall incorporate a text “shall” statement for the requirement in question,
4. Each requirement shall incorporate a rationale statement that provides context and additional background on the purpose of the requirement,
5. Each requirement shall include traceability between the Offeror proposed requirements and commercial objectives (and maintains explicit traces to parent NASA-unique requirements),
6. Each requirement shall include a verification method,
7. Each requirement shall include a statement describing the verification approach.

Requirements may also include additional supporting information as deemed needed by the Contractor.

Where needed, the requirement set shall include a summary of open “to-be-determined (TBD)” and “to-be-resolved (TBR)” and some description of the forward work plans needed to address them (or include other Contractor nomenclature for “open” values in the requirement set).

The full set of MTN System Requirements is comprised of [a] the NASA-unique requirements, MTN-402-REQ-0001 [b] the Contractor defined set of requirements that will form a complete set of spacecraft functional, performance, interoperability, and safety and mission assurance requirements.

Format: Delivery of data in electronic MS Excel file and native requirement database (DOORS, etc.) format

End of Data Requirements Description DRD-SE-01

Released Version

5.12 Data Requirements Description DRD-SE-02 (System Specification)

Program:	MTN	DRD No.:	DRD-SE-02
Data Type:	2	Date Revised:	Initial Release
Pages:	2		

Title: System Specification

Description/Use: The System Specification, also called an end item specification, documents a system and subsystem design specification derived from DRD-SE-01, System Requirements.

Initial Submission: Baseline by the end of Preliminary Design Phase

Submission Frequency: Updated at major milestones as identified in DRD-PM-02 (Integrated Master Schedule).

Remarks:

Contents:

The MTN System Specification shall document a description of the MTN Flight System and its interfaces, specify the MTN performance, specify requirements for design and construction, and establish the verification requirements. The MTN System Specification shall be responsive to DRD-SE-01, System Requirements. The Specification shall address, at minimum, the following topics, as appropriate:

Overview

1. Subsystem/Assembly Description and Characteristics
2. Subsystem/Assembly Block Diagram(s)

Requirements

1. Performance Requirements
2. Performance Budget sub-allocations derived from system budgets
3. Operational Requirements
4. Command and Telemetry Requirements
5. Fault Detection and Correction Requirements
6. Software Requirements

Resources

1. Power allocation, average, peak and transient curves
2. Mass Allocation/ Volume Allocation

Reliability

1. Reliability requirements
2. Redundancy description

3. Life characteristics, including consumables and life-limited items (mechanisms, memory degradation, valve cycles, etc.)

4. Single Point Failures

Verification

1. Testability requirements
2. Verification method requirement
3. Environmental test and analysis requirements.

Interfaces

1. Interfaces to other subsystems, including end-circuits and pin-outs
2. System Test Interfaces
3. Interfaces external to the spacecraft
4. Interfaces to Software

Environments

1. Pressure, Temperature, Vibration, Shock, Radiation, Magnetics, EMI, EMC, Grounding and Electrical Bonding requirements
2. Ground environments requirements

Design and Construction

1. Parts, Materials, Processes requirements
2. Contamination Control requirements
3. Magnetic Control requirements
4. Self-Compatibility

Safety

1. Personnel safety requirements
2. System safety requirements Packaging and Handling
3. Cleanliness, protection, cushioning and transport, storage

The contractor may provide the system specification in multiple volumes, as appropriate to their standard practices.

Format: Contractor's native format is acceptable.

End of Data Requirements Description DRD-SE-02

5.13 Data Requirements Description DRD-SE-03 (Interface Control Documents and Drawings)

Program:	MTN	DRD No.:	DRD-SE-03
Data Type:	2	Date Revised:	Initial Release
Pages:	1		

Title: Interface Control Documents and Drawings

Description/Use: To provide configuration controlled documentation in the form of drawings and/or written records to identify necessary design definitions between one or more systems, elements, subsystems, or computer software configuration items (CSCIs) to provide control of and ensure a compatible interface.

Initial Submission: Baseline by the end of Preliminary Design Phase

Submission Frequency: Final by the end of Detailed Design Phase

Remarks:

Contents:

The ICDs shall address the following class of interfaces*:

1. Physical – Interfaces involving physical mating and spatial relationships between interconnecting parts of interfacing end items, including clearance envelopes established to avoid interferences and to permit access.
2. Functional – Interfaces involving the interaction or influence of conditions imposed by one Component End Item (CEI) Interface or higher level assembly upon another or by external sources such as fluids, thermal, electrical, environmental, data, and loads.
3. Procedural – Interfaces involving critical sequence of events occurring in assembly, disassembly, alignment, service operations, and computer programs.

*The Contractor is encouraged to make use of existing document formats that meet the intent of this DRD.

Format: Contractor's native format is acceptable.

End of Data Requirements Description DRD-SE-03

5.14 Data Requirements Description DRD-SE-04 (Hardware Deliverable List)

Program:	MTN	DRD No.:	DRD-SE-04
Data Type:	3	Date Revised:	Initial Release
Pages:	1		

Title: Hardware Deliverable List

Description/Use: The Hardware Deliverable Lists serves to document the hardware that will be delivered to NASA by the contractor in performance of this contract.

Initial Submission: Baseline 60 days after Authority to Proceed (ATP).

Submission Frequency: Final by the end of Preliminary Design Phase

Remarks:

Contents:

The hardware deliverable list will identify all hardware that will be delivered by the contractor to NASA throughout the execution of the contract. This shall include:

1. Spacecraft.
2. Ground support equipment
3. Ground/Compatibility test hardware
4. Flight spares
5. Ground spares

For each item, the hardware deliverable list shall identify at a minimum:

1. Part number
2. Serial number (if applicable)
3. Design Revision
4. Quantity
5. Description

Format: Contractor's native format is acceptable.

End of Data Requirements Description DRD-SE-04

5.15 Data Requirements Description DRD-SE-05 (Acceptance Data Package of Delivered Hardware)

Program:	MTN	DRD No.:	DRD-SE-05
Data Type:	1	Date Revised:	Initial Release
Pages:	2		

Title: Acceptance Data Package of Delivered Hardware

Description/Use: The purpose of the Acceptance Data Package is to provide the documentation needed by NASA to establish the acceptability of the hardware and software for its intended use. The data comprising the software portion of the Acceptance Data Package (ADP) supports software deliveries for the MTN Flight System. Specific data identifying and representing the state of the software products being delivered is provided to the receiving organization and subsequently maintained throughout the life of the MTN.

Acceptance data packages are expected for all items identified in the hardware deliverable list.

Initial Submission: Preliminary 20 working days prior to Pre-ship Review (for spacecraft)

Submission Frequency: Final 20 working days prior to Acceptance Review

Remarks:

Contents:

The contents of the package shall include, as a minimum, the following information, as applicable to the item being delivered:

1. Item Description
2. Index of Data
3. Waivers / Deviations
4. Operating Time/Cycle Numbers
5. Non-Conformance Reports
6. Pressure Vessel Data
7. Problem/failure reports with root cause and corrective actions
8. As-Built Engineering Drawings, in native file formats.
9. As-Built Final Assembly Drawings
10. As-Built EEE Parts and Materials Lists
11. Limited Life Items / Age Sensitive Items, including data for life used and remaining life and periodic calibration items
12. Nonstandard Calibration Data
13. As-Run Test Procedures and Test Reports for system level testing

14. Acceptance Verification Matrix and Evidence of Acceptance Completion
15. Photographic documentation of hardware (subsystem, harnesses, structures, flight system)
16. Mass Properties Report
17. Contamination Conformance documentation
18. Sparing policies
19. Notes/Comments

Format: Contractor's native format is acceptable.

End of Data Requirements Description DRD-SE-05

Released Version

5.16 Data Requirements Description DRD-SE-06 (Environmental Requirements Document)

Program:	MTN	DRD No.:	DRD-SE-06
Data Type:	2	Date Revised:	Initial Release
Pages:	2		

Title: Environmental Requirements Document

Description/Use: The Environmental Requirements Document defines the natural and operational environments MTN will encounter from transportation to the launch site to disposal.

Initial Submission: Baseline by the end of Preliminary Design Phase

Submission Frequency: Updates as required.

Remarks:

Contents:

The Contractor document shall contain, at a minimum, the following:

1. Natural Environments for:
 - a. Ground Processing Phases
 - i. Transportation Environments
 - ii. Integration/Launch Site Environments
 - b. Launch and Ascent Phases
 - i. In-Space Phases
 - ii. Total Dose
 - iii. Single Event Effects
 - iv. Meteoroid and Orbital Debris Environment
 - v. Earth Gravitational Field
 - vi. Martian Gravitational Field
 - vii. Thermal Environment for In-Space Hardware
 - viii. Solar Illumination Environment for In-Space Hardware
2. Operational Environments for:
 - a. Ground Processing Phases
 - i. Thermal Environments
 - ii. Electromagnetic Interference (EMI)/Electromagnetic Compatibility (EMC)
 - b. Launch and Ascent Phases
 - i. Spacecraft/Payload Design Loads
 - ii. Acoustics
 - iii. Vibration

- iv. Shock
- v. Thermal
- vi. Static Pressure
- vii. Launch and On-Orbit EMI/EMC
- c. For All In-Space Phases
 - i. Material Outgassing
 - ii. Spacecraft Charging
 - iii. Radiation
 - iv. Self-induced shock from mechanism launch-locks, deployments, etc.
- d. Spaceflight Cruise Phase
 - i. Distance from the Earth and Mars
 - ii. Induced Reaction Control System (RCS) Plume
 - iii. Flight Attitudes
 - iv. Thermal Environments
 - v. EMI/EMC
- e. Relay Operations Phase
 - i. Distance from the Earth and Mars
 - ii. Induced Loads
 - iii. Flight Attitude
 - iv. Thermal Environments
 - v. EMI/EMC

Format: Contractor's native format is acceptable.

End of Data Requirements Description DRD-SE-06

5.17 Data Requirements Description DRD-SE-07 (Design Data Book)

Program:	MTN	DRD No.:	DRD-SE-07
Data Type:	2	Date Revised:	Initial Release
Pages:	2		

Title: Design Data Book

Description/Use: The MTN Design Data Book (DDB) provides an incremental view of the design process, progress and results of the contractor's design. It will provide insight into the contractor's design activities as well as reference information to support the engineering and operations throughout the life of MTN. It will include an overview, key trades or analyses, spacecraft modes and fault protection, and physical configuration. The DDB compiles results for both the integrated spacecraft as well as spacecraft subsystems.

Initial Submission: Preliminary by the end of Preliminary Design Phase

Submission Frequency: Updated at major milestones as identified in DRD-PM-02 (Integrated Master Schedule). Final 20 working days prior to Acceptance Review

Remarks:

Contents:

The Design Data Book should include:

1. Overview of Design Configuration of MTN at end Design Cycle
 - a. Integrated System and subsystem overview descriptions, functionality, schematics and capabilities. Brief overview of what components are included in the flight system
 - b. Description of all requirements driving flight system and subsystem designs.
 - c. Description of the architectural degrees of freedom for the flight system and subsystems
 - d. Description of the key architectural guidelines or constraints for the flight system and subsystems including physical, functional, information, software.
 - e. Diagram and describe all major physical configurations of the spacecraft (stowed, deployed, on-orbit)
 - f. Diagram and describe all system-level coordinate frames
 - g. List any subsystem coordinate frames
 - h. Show all external faces of the spacecraft and main dimensions including component locations. Depict Fields-of-View where relevant
2. Overview of the Design Metrics (included but not limited to):
 - a. Master Equipment List (MEL)

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- b. Power Equipment List (PEL)
 - c. Link Budget
 - d. Communications Coverage
 - e. Spacecraft margins for the MTN system and subsystems
 3. Overview of the Spacecraft modes, states, behaviors and fault protection.
 - a. Describe and diagram all spacecraft system-level state and sub- states. Include relevant or affected subsystem states and how they relate to the system state.
 - b. Identify primary spacecraft behaviors (complete description can be included in the FSW area)
 - c. Depict state transition diagrams and allowable and unallowable state transitions
 - d. Describe and show behavior relationships and priorities
 - e. Describe overview of fault protection implementation including system responses, monitors, thresholds and intended mitigations.
 - f. Describe and diagram redundancy and any cross-strapping
 - g. Describe fault protection design philosophy between hardware and software functions and implementations.
 4. Key Trade Studies or Key Analyses
 - a. Top-level description or summary of any trade studies or analysis conducted resulting in the implemented options. Include process used for each trade or analysis.
 - b. References to documentation or other DRDs containing details of trade studies or analyses. (Design Data Book is intended to be the collection of assessments describing the resulting implementation)
 - c. List of any in-process or known future trade studies or analyses.
 - d. Key decisions made as a result of trade studies or analyses

Format: Contractor's native format is acceptable.

End of Data Requirements Description DRD-SE-07

5.18 Data Requirements Description DRD-SE-08 (Spacecraft Performance and System Resource Budget Reports)

Program:	MTN	DRD No.:	DRD-SE-08
Data Type:	3	Date Revised:	Initial Release
Pages:	1		

Title: Spacecraft Performance and System Resource Budget Reports

Description/Use: The Spacecraft Performance and System Resource Budget and Margin Reports shall document all relevant technical resource budgets to include mass properties, physical volume, power usage, delta-v, pointing, communications link.

Initial Submission: Baseline by the end of Preliminary Design Phase

Submission Frequency: Updated at major milestones as identified in DRD-PM-02 (Integrated Master Schedule). Final 20 working days prior to Acceptance Review

Remarks:

Contents:

The report is based upon calculated values and is updated as calculations are revised and actual measured data becomes available. The Spacecraft Performance and System Resource Budget report contains the following.

1. Mass properties for the MTN system and subsystems including mass, center of gravity, moments of inertia, products of inertia, principal axis misalignment, appropriate margins and tolerances.
2. Overall MTN Spacecraft dimensions.
3. Performance budgets for data throughput, data latency, CPU usage, and memory utilization.
4. Spacecraft bus power usage summary for the MTN system and subsystem for the various phases of the mission. This summary should also demonstrate power and energy storage throughout the mission through EOL, including power margin estimates.
5. Spacecraft margins for the MTN system and subsystems.
6. Communications coverage and Summary Link Budget cover link availability considering vehicle trajectory, antenna patterns, structural blockage, planet blockage, and link margins.
7. Summary of all changes incorporated into the spacecraft bus Spacecraft Performance and System Resource Budget report since the last delivery, including associated rationale.
8. List of pending and potential changes.

Format: Contractor's native format is acceptable.

End of Data Requirements Description DRD-SE-08

Released Version

5.19 Data Requirements Description DRD-SE-09 (Assembly, Integration and Test Plan)

Program:	MTN	DRD No.:	DRD-SE-09
Data Type:	2	Date Revised:	Initial Release
Pages:	2		

Title: Assembly, Integration and Test (AI&T) Plan

Description/Use: The Assembly, Integration and Test (AI&T) Plan documents the contractor's system level AI&T methodology and processes. It also documents the detailed AI&T activities needed to prove conformity with the technical specifications and perform design certification and end item acceptance.

Initial Submission: Baseline by the end of Preliminary Design Phase

Submission Frequency: Updated at major milestones as identified in DRD-PM-02 (Integrated Master Schedule).

Remarks:

Contents:

The Contractor's plan* shall contain, at a minimum the following:

1. Overall MTN System Level Test Program requirements and objectives.
2. Description of the individual tests to be performed, their configuration (including ground systems), test objectives, support equipment, special equipment, and requirements addressed by the test.
3. Number of operating hours required to be executed pre-launch on all flight electronics.
4. Description of the plan for safe conduct of operations (flight hardware and personnel).
5. Flight hardware contamination and environmental control requirements and implementation.
6. Flight hardware security requirements and implementation.
7. Requirements for handling deviations, exceptions, omissions and discrepancies during tests.
8. Listing of additional plans and procedures to be written in support of this plan for all contractor-defined key system and subsystem level tests, including but not limited to the following:
 - a. Vibration/Acoustic
 - b. Thermal Vacuum
 - c. EMI/EMC
 - d. Power Quality
 - e. System Level End-to-End Validation (Ground to MTN, Comm links)
 - f. Functional

- g. System Level Interfaces to External Emulators/Simulators
9. Transportation plan for all flight hardware moves.
 10. Plan for support equipment certification.
 11. Plan for control of Flight System configuration during AI&T.
 12. Use of quality assurance.
 13. Control and archival of controlled records (including as-run test procedures and data)
 14. Network schedule including integration of flight subsystem feeder schedules (including GFE).
 15. Identification of receivables and deliverables for all flight hardware, flight software, flight sequences, ground support equipment, and documentation (baseline).
 16. Process for ensuring that lower level of assemblies are ready for integration into the system AIT flow (e.g., subsystem acceptance approach).
 17. Integration, Test, and Handling constraints.
 18. Identification of Risks and associated mitigations.
 19. Plans for testing, and (3) mission scenarios using the Flight Test Bed(s) and the Flight System.
 20. Plans for activities and testing at the launch site, including propellant loading, post-shipment tests of the spacecraft(s), final closeouts, and any combined testing with the launch vehicle.
 21. Identification of NASA provided Ground Support Equipment (mechanical, electrical) and software needed to verify that the flight system meets all technical requirements.
 22. Description of all supporting hardware and resources
 23. Identification of Government provided facilities or support needed to verify that the flight system meets all technical requirements.

*The Contractor is encouraged to make use of existing standardized plans, procedures and related products that meet the intent of this DRD. It is permissible to use multiple contractor standard plans and procedures to meet the intent of this DRD (rather than creating a single document).

Format: Contractor's native format is acceptable.

End of Data Requirements Description DRD-SE-09

5.20 Data Requirements Description DRD-SE-10 (Verification & Validation Plan)

Program:	MTN	DRD No.:	DRD-SE-10
Data Type:	2	Date Revised:	Initial Release
Pages:	2		

Title: Verification & Validation Plan

Description/Use: The Verification and Validation Plan documents the verification methodology and processes, as well as the validation approach and processes to be used by the Contractor. It also documents the detailed verification activities needed in order to prove conformity with the technical specifications and the validation activities needed to confirm that MTN will operate as expected in the intended environment.

Initial Submission: Baseline by the end of Preliminary Design Phase

Submission Frequency: Updated at major milestones as identified in DRD-PM-02 (Integrated Master Schedule).

Remarks:

Contents:

The Verification and Validation Plan shall include, at a minimum*:

1. Overview of the Contractor verification and validation program.
2. Verification matrix that clearly documents the verification method or methods employed for each system level requirement and each allocated requirement captured in DRD-SE-01, System Requirements, as well as verification closure data type. Detailed verification descriptions for each DRD-SE-01 requirement can be used as draft VCNs (see DRD-SE-13).
3. Description of the Verification and Validation Process:
 - a. Philosophy
 - b. Approach
 - c. V&V processes
 - d. V&V planning
 - e. V&V tracking
 - f. Close-out process
4. Incompressible Test List
5. Description of the V&V Phases.
6. Description of hardware and software verification environments and validation environments that are to be established for the project.

7. Description of how software and hardware verification and validation activities will be mapped to flight hardware and support environments.
8. Definition of verification methods.
9. Detailed time-correlated sequence of verification activities and associated requirement verifications that occur.
10. Description of all supporting hardware and resources.
11. Certification plans/processes for all ground support equipment (GSE).
12. For each validation activity identified (i.e. major test or demonstration event), provide a detailed validation task description including: identification of specific objective, capability, or function being validated, method of validation, measures of performance or effectiveness used for validation, and enabling products and support resources (facilities, models, software, simulators, etc.) required to perform the activity.

*The Contractor is encouraged to make use of existing document formats that meet the intent of this DRD.

Format: Contractor's native format is acceptable.

End of Data Requirements Description DRD-SE-10

5.21 Data Requirements Description DRD-SE-11 (Test Plans and Procedures)

Program:	MTN	DRD No.:	DRD-SE-11
Data Type:	3	Date Revised:	Initial Release
Pages:	2		

Title: Test Plans and Procedures

Description/Use: Test plans are used to develop and document overall approaches to conducting tests. Detailed “how-to” instructions are generally contained in test procedures or work instructions.

Initial Submission: Baseline delivered with Test Readiness Review Package

Submission Frequency: Final within 10 working days of test completion.

Remarks:

Contents:

The Subcontractor test plans* should contain the following:

1. Overall Test requirements, objectives, and purpose of the plan.
2. Description of the individual tests to be performed, their configuration (including test articles, facilities, and ground systems), test objectives, success criteria (for requirements verification), support equipment, special equipment, requirements addressed by the test, data collected, test duration, test levels and included margin, and test factor.
3. Description of the plan for safe conduct of operations (flight hardware and personnel).
4. Flight hardware contamination and environmental control requirements and implementation.
5. Flight hardware security requirements and implementation.
6. Requirements for handling deviations, exceptions, omissions and discrepancies during tests.
7. Transportation plan for all flight moves.
8. Plan for support equipment certification.
9. Use of quality assurance.
10. Identification of Risks and associated mitigations.
11. Control and archival of controlled records (including as-run test procedures and data).
12. Test schedule, planned test reviews and documentation.
13. Identification of receivables and deliverables for all flight hardware, flight software, flight sequences, ground support equipment, and documentation (baseline).
14. Integration, Test, and Handling constraints.
15. Identification of NASA provided Ground Support Equipment (mechanical, electrical) and software needed to verify that the flight system meets all technical requirements.

The Subcontractor test procedures* should contain the following:

1. Description of the test configuration including test hardware, supporting hardware with calibration data, and software configuration if applicable.
2. Description of the test facility.
3. Description of the key test personnel and organizations responsible for the test.
4. Description of test setup.
5. Detailed “how-to” instructions with appropriate sign-offs.
6. Description of failure and anomaly protocol.
7. Description of post-test break-of-configuration process for flight hardware.

*The Contractor is encouraged to make use of existing standardized plans, procedures and related products that meet the intent of this DRD. It is permissible to use multiple contractor standard plans and procedures to meet the intent of this DRD (rather than creating a single document). Test plans should be developed commensurate with the specific testing needs. The degree of planning required may be tailored to scope and complexity of the test in question.

Format: Contractor’s native format is acceptable.

End of Data Requirements Description DRD-SE-11

5.22 Data Requirements Description DRD-SE-12 (Test Reports)

Program:	MTN	DRD No.:	DRD-SE-12
Data Type:	3	Date Revised:	Initial Release
Pages:	1		

Title: Test Reports

Description/Use: Test reports document results obtained during assembly, subsystem and system testing, as well as flight system-level integration, environmental and performance testing.

Initial Submission: Within 20 working days of test completion

Submission Frequency:

Remarks:

Contents:

At a minimum, each Test Report* requires the following information:

1. Description of the test configuration, instrumentation, and test sequence,
2. Compilation of the test data, trend data, and results,
3. Explicit reference to verification requirements that the test will be used to close,
4. A list and description of specific deviations/changes from the approved test plan or test procedures,
5. Copies of all non-conformance reports, MRB actions or other documents, which define problems/changes/deviations from the approved test plans or test procedures,
6. Reference to as-run test procedures and summary test data recorded during test,**
7. Copy of logs with mate/demates on all flight-interface connectors and flight flanges and element to element structural interfaces including flight support equipment.

*The Contractor is encouraged to make use of existing standardized report formats and related products that meet the intent of this DRD.

** The Contractor shall retain all test records and make them available upon request.

Format: Contractor's native format is acceptable.

End of Data Requirements Description DRD-SE-12

5.23 Data Requirements Description DRD-SE-13 (Verification Closure Notices and Validation Closure Notices/Requirements Verification Traceability Matrix)

Program:	MTN	DRD No.:	DRD-SE-13
Data Type:	2	Date Revised:	Initial Release
Pages:	2		

Title: Verification Closure Notices and Validation Closure Notices/Requirements Verification Traceability Matrix

Description/Use: Verification closure notices (VCNs) shall be provided for each requirement, including safety hazard requirements, identified in the verification matrix. The VCN contains information identifying the requirement and a brief description of the verification results along with a list of supporting documentation providing objective evidence of the requirement verification.

The Requirements Verification Traceability Matrix (RVTM) links requirements throughout the verification and validation process and reports the results of the verification activities by identifying and correlating the Verification Closure Notices against the requirements

Initial Submission: Preliminary matrix delivered during preliminary design phase. Preliminary notices within 5 working days after event.

Submission Frequency: Updated matrix at major milestones as identified in DRD-PM-02 (Integrated Master Schedule). Final matrix 20 working days before pre-ship review. Final notices within 20 working days after event.

Remarks:

Contents:

Verification closure notices shall contain, at a minimum:

1. Requirement number(s)
2. Traceability to NASA requirement (if applicable)
3. Details of the specific verification activity
4. Any non-conformances or deviations from the flow-down requirements.
5. Supporting documentation including supporting technical data
6. The verification method and results (pass/fail)
7. Details of anomalies found and their resolution, including plans to perform the verification activity again if necessary or recommendation to process a deviation/waiver request
8. Summary of the results of the verification activity, tests, demonstrations, inspections, and/or analyses performed with any anomalies noted.

9. A recommendation to certify the requirement as closed, re-verify the requirement, or issue a waiver.
10. Signature by responsible authority verifying requirement closure

The Requirements Verification Traceability Matrix shall include, as a minimum:

1. Requirement ID
2. Risks
3. Requirement Description
4. Trace to design specification and System parent MTN requirement(s).
5. Links to a single data folder that contain copies of all compliance data for each specific requirement (i.e. folder should not contain additional links). Note: For any data that cannot be stored within the specified folder and must be accessible, the folder will contain a file describing the data which includes the date posted, data files, and revisions.
6. Identification of any non-conformances (e.g. waivers, deviations, discrepancy report) against the requirement(s).
7. Include unique identifiers that correlate to each verification method applied to satisfy a requirement.
8. Include success criteria for each method to address compliance with the requirement.

Format: Contractor's native format is acceptable.

End of Data Requirements Description DRD-SE-13

5.24 Data Requirements Description DRD-SE-14 (Drawing and Document Trees)

Program:	MTN	DRD No.:	DRD-SE-14
Data Type:	3	Date Revised:	Initial Release
Pages:	1		

Title: Drawing and Document Trees

Description/Use: To provide a categorized listing of drawings and documents required for development, design, manufacturing, verification, validation, qualification, and design evolution of a Configuration Item (CI) or Computer Software Configuration Item (CSCI).

Initial Submission: Preliminary due during preliminary design phase.

Submission Frequency: Baseline due during detailed design phase.

Remarks:

Contents:

The document tree shall consist of an indentured or hierarchical breakdown listing of all specifications, plans, reports, or other documents applicable to a configuration item.

The drawing tree shall consist of an indentured or hierarchical breakdown listing of all drawings applicable to a configuration item.

Format: Contractor's native format.

End of Data Requirements Description DRD-SE-14

5.25 Data Requirements Description DRD-SE-15 (Engineering Drawings, Models, and Associated Lists)

Program:	MTN	DRD No.:	DRD-SE-15
Data Type:	3	Date Revised:	Initial Release
Pages:	2		

Title: Engineering Drawings, Models, and Associated Lists

Description/Use: To provide engineering data defining the design to the extent required to support manufacturing, testing prior to delivery, integration into the launch vehicle, any remaining pad processing activities, any remaining testing from point of delivery through launch, logistics support including required spare parts. Engineering drawings and associated lists shall be sufficient to depict the detailed configuration of all system, subsystem, and component levels. The corresponding 2D and 3D CAD models shall be submitted in accordance with DRD Engineering CAD Models and Data.

Initial Submission: 30% complete by the end of the Preliminary Design Phase; 90% complete by the end of the Detailed Design Phase; Final complete by the end of the Fabrication, Build, Assembly & Test Phase.

Submission Frequency: Released drawings shall be submitted prior to the manufacture of any qualification test or flight hardware. Update released drawings as a required for significant design changes and each Acceptance Data Package

Remarks:

Contents:

1. Drawings and parts lists shall be provided for any assemblies, subassemblies, parts, installations, or kits which define the product(s) developed under this contract, including flight hardware and ground support equipment. CAD Models delivered shall be traceable to the specific version of drawings delivered.
2. Geometric Dimensioning and Tolerancing shall be implemented consistent with the Contractor's standards and practices. Engineering drawings and associated lists of end items, elements and/or all components and assemblies shall be provided to define the details necessary for the manufacture, test, inspection, operations and logistic support of the system. This definition shall:
 - a. Reflect the end-product.
 - b. Provide the engineering data for logistics support products.
 - c. Provide the necessary data to permit manufacture and/or acquisition of items identical to the original item(s).

- d. Document directly or by reference the following:
- i. Details of unique processes (i.e., not published or generally available to industry) when essential to design and manufacture.
 - ii. Performance ratings when applicable.
 - iii. Dimensional and tolerance data (Geometric Dimensioning and Tolerancing (GD&T) shall be required between all external and major internal interfaces).
 - iv. Critical manufacturing processes and assembly sequences, and rigging procedures.
 - v. Diagrams and schematics not provided per other DRDs.
 - vi. Mechanical and electrical connections.
 - vii. Physical characteristics, including form and finish.
 - viii. Details of material identification, including heat treatment and protective coatings.
 - ix. Inspection, test, and evaluation criteria.
 - x. Equipment calibration requirements.
 - xi. Quality assurance requirements.
 - xii. Hardware marking requirements.
 - xiii. Requirements for reliability, maintainability, environmental conditions, shock, and vibration testing and other operational or functional tests.
 - xiv. Drawing tree or indentured parts list (per DRD-SE-14, Drawing Tree).

Format: Drawings shall be delivered in PDF format. The corresponding 2D and 3D CAD models shall be submitted in Contractor's native format.

End of Data Requirements Description DRD-SE-15

5.26 Data Requirements Description DRD-SE-16 (Design and Construction Standards Utilized)

Program:	MTN	DRD No.:	DRD-SE-16
Data Type:	3	Date Revised:	Initial Release
Pages:	1		

Title: Design and Construction Standards Utilized

Description/Use: Describe how the industry utilized design and construction standards selected for the MTN take full advantage of the demonstrated reliability benefits of commercial spacecraft. This DRD allows the Government to assess potential gaps between NASA and industry standards.

Initial Submission: Baseline upon completion of Preliminary Design Phase

Submission Frequency: Final upon completion of Fabrication, Build, Assembly & Test Phase

Remarks:

Contents:

1. Provide a description of the approach to the identification and application of design and construction standards and specifications to be used for the development of the MTN as required to take full advantage of the demonstrated reliability benefits of commercial spacecraft.
2. Specific content includes:
 - a. A complete list of denoting the document unique identifier, document name, subject area and applicable version or release date to define the Contractor's standards and specifications to be applied to the development of the MTN.
 - b. For each standard, justification for the suitability of using industry non-equivalent standards and specifications including relevant data on past usage and other supplemental information that will allow NASA to adequately assess the risk associated with their usage.
 - c. Copies of the Contractor's design and construction standards and specifications not found in the public domain shall be provided electronically.
 - d. With the final submittal, identify all deviations and waivers against the applicable design and construction standards and submit as part of DRD-SMA-01, Request for Deviation or Waiver.

Format: Contractor's native format.

End of Data Requirements Description DRD-SE-16

5.27 Data Requirements Description DRD-SE-17 (Structural Analysis Reports)

Program:	MTN	DRD No.:	DRD-SE-17
Data Type:	3	Date Revised:	Initial Release
Pages:	2		

Title: Structural Analysis Reports

Description/Use: The stress analysis report documents the strength and life integrity of the structure and provides documentation for the structural certification of the hardware. The stress analysis report will establish the strength and life capability of the hardware, provide a method to evaluate proposed changes to the design loads and environment, and will provide data to resolve manufacturing, salvage, and field maintenance problems.

Initial Submission: Baseline upon completion of Preliminary Design Phase

Submission Frequency: Final upon completion of Fabrication, Build, Assembly & Test Phase

Remarks:

Contents:

The stress analysis report shall contain the following elements:

1. A general description of the system being analyzed.
2. A discussion of the methods and assumptions used in the analysis.
3. A margin of safety summary table that lists the lowest margin of safety for each part in the system and a cross-referenced page number where the margin calculation is performed.
4. Material property references and allowables summary.
5. Summary of load conditions and references.

The stress analysis shall be able to be read and understood without having to secure drawings. A sketch of the structure being analyzed shall be provided that will describe

1. What the part is and what it looks like
2. Where the part is located in the assembly or installation
3. Where the external loads are applied
4. Where the reaction loads are located
5. Drawing numbers for all parts unless already identified in the analysis
6. Dimensions and tolerances if applicable

Finite element analyses information shall include a description of the analyses with assumptions, applicable geometry, dimensions, loads, other boundary conditions, annotated input data file(s), plots of model geometry, results of model checks, and results. All information to recreate the analysis shall be provided.

The calculations documented in this report shall be formally checked and verified by the responsible engineering organization, with the checker's initials on each page of the report. Analyses must be checked prior to flight drawing release.

Format: Contractor's native format.

End of Data Requirements Description DRD-SE-17

Released Version

5.28 Data Requirements Description DRD-SE-18 (Thermal Analysis Reports)

Program:	MTN	DRD No.:	DRD-SE-18
Data Type:	3	Date Revised:	Initial Release
Pages:	1		

Title: Thermal Analysis Reports

Description/Use: Thermal Analysis Reports document the results of analyses performed in support of all phases of hardware development and use, and requirement verification.

Initial Submission: Baseline upon completion of Preliminary Design Phase

Submission Frequency: Final upon completion of Detailed Design Phase

Remarks:

Contents:

All analysis reports shall address, at a minimum, all topics listed below. The Contractor shall standardize an analysis report format.

1. Cover page with title, document number for traceability, relevant part and serial numbers (if applicable), analysis cycle, indication of purpose
2. Executive Summary
3. Objectives/Motivation for the analysis
4. Study Requirements and Constraints
5. Assumptions and ground rules
6. Initializing Data
7. Figures of merit
8. Analysis tools, tests, and Models used
9. Modeling technique
10. Internal Verification and Validation of the results (if applicable)
11. Uncertainty of the results
12. Description of sensitivities
13. Detailed results
14. Key findings/recommendations

Format: Contractor's native format.

End of Data Requirements Description DRD-SE-18

5.29 Data Requirements Description DRD-SE-19 (Power Analysis Report and Battery Usage Assessment)

Program:	MTN	DRD No.:	DRD-SE-19
Data Type:	3	Date Revised:	Initial Release
Pages:	1		

Title: Power Analysis Report and Battery Usage Assessment

Description/Use: Power Analysis Reports document the results of analyses performed in support of all phases of hardware development and use, and requirement verification. The power analysis report documents power margins and battery usage inclusive of depth of discharge, cycles, and overall life estimates.

Initial Submission: Baseline upon completion of Preliminary Design Phase

Submission Frequency: Final upon completion of Detailed Design Phase

Remarks:

Contents:

All analysis reports shall address, at a minimum, all topics listed below.

1. Cover page with title, document number for traceability, relevant part and serial numbers (if applicable), analysis cycle, indication of purpose
2. Executive Summary
3. Objectives/Motivation for the analysis
4. Study Requirements and Constraints
5. Assumptions and ground rules
6. Figures of merit
7. Analysis tools, tests, and Models used
8. Modeling technique
9. Internal Verification and Validation of the results (if applicable)
10. Uncertainty of the results
11. Description of sensitivities
12. Detailed results
13. Key findings/recommendations

Format: Contractor's native format.

End of Data Requirements Description DRD-SE-19

5.30 Data Requirements Description DRD-SE-20 (Electromagnetic Interference (EMI) Electromagnetic Compatibility (EMC) Control Plan)

Program:	MTN	DRD No.:	DRD-SE-20
Data Type:	3	Date Revised:	Initial Release
Pages:	1		

Title: Electromagnetic Interference (EMI) Electromagnetic Compatibility (EMC) Control Plan

Description/Use: The EMI/EMC control plan documents the strategy used to control EMI/EMC.

Initial Submission: Baseline upon completion of Preliminary Design Phase

Submission Frequency: Final 20 working days prior to pre-ship review.

Remarks:

Contents:

The EMI/EMC control plan shall describe the design and approach to comply with EMI/EMC requirements. The plan shall address all anticipated environments, mitigations, conducted/radiated emissions, conducted/radiated susceptibilities, and self-compatibility for all operational modes including multiple communications payloads operating simultaneously in their “worst case” configuration. Considerations include harnessing, bonding, and grounding.

Format: Contractor’s native format.

End of Data Requirements Description DRD-SE-20

5.31 Data Requirements Description DRD-SE-21 (Worst Case Analysis)

Program:	MTN	DRD No.:	DRD-SE-21
Data Type:	3	Date Revised:	Initial Release
Pages:	2		

Title: Worst Case Analysis

Description/Use: The Worst Case Analysis (WCA) demonstrates design margins in electronic and electrical circuits. This analysis assesses the effect of environmental stresses and component variability on system performance requirements.

Initial Submission: Baseline upon completion of the Detailed Design Phase

Submission Frequency: Final 20 working days prior to pre-ship review.

Remarks:

Contents:

The WCA, at a minimum, shall:

1. Derive part parameter variations for the intended environments and planned duration of the specified mission.
2. Combine part parameter variations with the initial tolerances of the parts as procured to produce a worst case part variation database for the mission.
3. Typical part parameters are voltage, current, resistance, impedance, inductance, and temperature.
4. Apply classical circuit analysis techniques to determine if each circuit and each assembly meets its specified performance outputs over the most extreme but realizable combinations of part performance variations.
5. Be facilitated by reducing complex circuits into smaller functional blocks. By using this approach the analysis becomes more manageable, so both the analysts and the reviewers are aided. When a circuit is reduced to these functional blocks, performance requirements for each block need to be established.
6. State all assumptions (clearly identified) including boundary conditions. These may include conservative simplifying assumptions that make the analysis tractable and/or more cost effective. Boundary conditions may include physical or functional interfaces with other elements or hardware or may involve a limitation on the number of functions analyzed. Where functions are not analyzed, the rationale for that decision must be documented.

The WCA report, at a minimum, shall:

7. Show compliance with all performance requirements, both on the functional block level and at the circuit level.
8. Note explicitly any deviations from the performance requirements.

9. Explain any proposed solutions or deviations.

Format: Contractor's native format.

End of Data Requirements Description DRD-SE-21

Released Version

5.32 Data Requirements Description DRD-SW-01 (Software Requirements Specification)

Program:	MTN	DRD No.:	DRD-SW-01
Data Type:	3	Date Revised:	Initial Release
Pages:	2		

Title: Software Requirements Specification (SRS)

Description/Use: The Software Requirements Specification is used to define and record the software requirements to be met by a computer software configuration item (CSCI) in sufficient detail to complete the software development effort. This document includes the methods to ensure that each requirement has been met and forms the basis for design, development, and certification of a CSCI.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Final 20 working days prior to pre-ship review

Remarks:

Contents:

The Software Requirements Specification shall include at a minimum:

1. System overview
2. CSCI requirements
 - a. Functional requirements
 - i. Specification of nominal behavior in terms of applicable parameters, such as sequencing, accuracy, capacities (how much/ how many), task priorities, continuous operation requirements, and allowable deviations based on operating conditions.
 - ii. Identification of unexpected, not allowed, or “out-of-bounds” conditions, requirements for error handling, and any provisions needed to ensure continuity of operations during emergencies.
 - iii. Definition of all required operating modes and states.
 - b. Interface requirements
 - i. External interface requirements
 - ii. Internal interface requirements
 - c. Safety requirements
 - d. Performance and timing requirements
 - e. Security and privacy requirements
 - f. Environment requirements
 - g. Computer resource requirements
 - i. Computer hardware resource use requirements
 - ii. Computer software use requirements

- iii. Computer communications requirements
 - h. Software quality characteristics
- 3. Supporting requirements rationale
- 4. Verification method(s)
- 5. Requirements traceability

Format: Contractor's native format.

End of Data Requirements Description DRD-SW-01

Released Version

5.33 Data Requirements Description DRD-SW-02 (Master Data Dictionary)

Program:	MTN	DRD No.:	DRD-SW-02
Data Type:	3	Date Revised:	Initial Release
Pages:	2		

Title: Master Data Dictionary

Description/Use: The Master Data Dictionary is used to define, record, and manage all commands and data items/elements that interface with the core avionics software and hardware, the subsystem specific software, and the ground systems. The content for this DRD will be developed as a joint NASA-contractor effort. The Master Data Dictionary will consist of a document, as well as a sortable electronic database.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Final 20 working days prior to pre-ship review

Remarks:

Contents:

The Master Data Dictionary shall consist of a document and a sortable database for use by multiple end-user functions. The document shall contain:

1. Description of the general commanding process
2. Command format(s)
3. Description of the general on board telemetry process including downlink type (packet, frame, etc.)
4. Telemetry format(s)
5. APID (Application Program Interface Division/Definition) definitions and descriptions.
6. Channelization data [e.g., bus mapping, wiring mapping, and end-to-end hardware channelization (from sensor/transducer, through signal conditioning, and multiplexing, to final destination of measurement)]

The sortable database shall contain:

1. All commands, including the following attributes for each command:
 - a. Command name
 - b. Command op code
 - c. Command parameters
 - d. Command function/description
 - e. Applicable subsystems
 - f. Vehicle modes and software states in which the command is valid
 - g. Allowable command source(s) in each of the valid modes/states

-
- h. Whether this command is restricted (Restricted commands if improperly executed could result in permanent damage to the spacecraft, or other undesirable operational impacts)
 2. All data, including the following attributes (at a minimum) for each: NOTE: If any attribute values differ by vehicle modes/software states, specify the value in each mode/state.
 - a. Data identifier
 - b. Subsystem and/or component to which the data belongs
 - c. Description
 - d. Type of telemetry (analog, discrete)
 - e. Engineering units
 - f. Polynomial coefficients used to convert data to engineering units.
 - g. Limits - maximum and minimum values, including off-scale high/low limits and Caution & Warning levels (if applicable)
 - h. Rate the data is collected
 - i. Rate the data is sent as telemetry
 - j. Telemetry recipient(s)
 - k. Data storage rate
 - l. Data configuration/reconfiguration status
 - m. Duration the data is stored on MTN System/deletion rules

Format: Contractor's native format.

End of Data Requirements Description DRD-SW-02

5.34 Data Requirements Description DRD-SW-03 (Software Management and Development Plan)

Program:	MTN	DRD No.:	DRD-SW-03
Data Type:	3	Date Revised:	Initial Release
Pages:	2		

Title: Software Management and Development Plan

Description/Use: The Software Management and Development Plan describes the approach for managing and conducting the software development effort. The plan provides insight into and is a tool for monitoring the processes and methods to be followed for software development, and the approach to be followed for each activity, based on project schedules, organization, and resources. This Plan details the software systems, project documentation, project schedules, resource requirements and constraints, and general and detailed software management and development activities

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Final upon completion of Detailed Design Phase

Remarks:

Contents:

Note that elements of this plan may be provided in separate documents provided the elements are clearly identified in terms of location and that the same delivery and approval requirements apply to those documents. The plan shall contain:

1. System overview, including identification of software components, their functions and characteristics, and their interfaces, planning assumptions and constraints
2. Software project organization and management:
 - a. Software organizational structure showing authority and responsibility of each unit and relationships to organizational units outside of software (e.g., Safety and Mission Assurance, Systems Engineering).
 - b. Identification of the software components and/or activities each unit is responsible for.
 - c. Software Work Breakdown Structure (WBS)
3. Software configuration and change management
 - a. Description of the software configuration management process
 - b. Description of the software change control process
 - c. Change management
 - d. Release management
4. Software lifecycle and development planning

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- a. Software life cycle model(s) to be used, including software development phases and their sequence.
 - b. System and software reviews of deliverables as specified in the statement of work and where the reviews occur in the software life cycle model(s).
 - c. Key software deliveries and where they occur in the software life cycle model(s).
5. Software test planning
 - a. Test philosophy
 - b. Overall approach for the effective and efficient removal of software defects, including levels of test and success criteria for progressing from one level to the next
 6. Software risk management
 7. Safety-critical and mission-critical software
 8. Software security
 9. Identification of any open source software being used.
 10. Approval required by regulations, required certifications, propriety, usage, ownership, warranty and licensing rights.

Format: Contractor's native format.

End of Data Requirements Description DRD-SW-03

5.35 Data Requirements Description DRD-SW-04 (Software Design Description/Version Description Document)

Program:	MTN	DRD No.:	DRD-SW-04
Data Type:	3	Date Revised:	Initial Release
Pages:	1		

Title: Software Design Description/Version Description Document

Description/Use: To provide insight into the software architecture, components, and interface design, as the foundation for detailed design and development.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Final 20 working days prior to pre-ship review

Remarks:

Contents:

The Software Design Description shall include:

1. Design goals, philosophy, assumptions, constraints, methods, and tools
2. System-wide software architectural design, including external hardware and software interfaces
3. CSCI architectural design, including hardware and software interfaces
4. CSCI decomposition and interrelationship between components.
 - a. Description of how the software component satisfies the software requirements, including algorithms and functional decomposition
 - b. Software component input/output description
 - c. Static/architectural relationship of the software components
 - d. Concept of execution including data flow, control flow, and timing
 - e. Identification of safety critical design elements
 - f. Plans for software re-use
 - g. State diagrams
5. Traceability to software requirements

Format: Contractor's native format.

End of Data Requirements Description DRD-SW-04

5.36 Data Requirements Description DRD-SMA-01 (Request for Deviation or Waiver)

Program:	MTN	DRD No.:	DRD-SMA-01
Data Type:	1	Date Revised:	Initial Release
Pages:	2		

Title: Request for Deviation or Waiver

Description/Use: A deviation is a documented authorization releasing a program or project from meeting a requirement before the requirement is put under configuration control at the level the requirement will be implemented. A waiver is a documented authorization releasing a program or project from meeting a requirement after the requirement is put under configuration control at the level the requirement will be implemented.

Initial Submission: As Needed

Submission Frequency: As Needed

Remarks:

Contents:

Each waiver/deviation request shall contain the following as a minimum:

1. Identification of the request as a waiver request or a deviation.
2. Equipment name, part number, applicable serial numbers, function and criticality, and effectivity and flight effectivity if applicable.
3. Assessment of the impact to safety, maintainability, reliability, performance, weight, interface requirements or basic objectives of program or contract requirements, including any of their related deliverable products (e.g. FMEA/CIL, Fault Tree Analysis, Hazard Analyses, etc.).
4. Number, title, and paragraph of affected program or contract documentation.
5. Identification of and cross-reference with, control documentation related to deviations/waivers.
6. Detailed description of discrepancy to include the differences between the design baseline and the requested configuration.
7. Rationale for acceptance and assessment of risk involved.
8. Effect of failure of the requested change.
9. Effects of delivery schedule (FMEA/CIL number, if applicable).
10. Reference to related deviation/waivers.
11. Technical justification supporting approval of the request.
12. The Contractor's reliability evaluation of the request.
13. Space for NASA review comments.
14. Approving signatures and date.

Format: Contractor's native format.

End of Data Requirements Description DRD-SMA-01

Released Version

5.37 Data Requirements Description DRD-SMA-02 (Safety and Mission Assurance (S&MA) Plan)

Program:	MTN	DRD No.:	DRD-SMA-02
Data Type:	2	Date Revised:	Initial Release
Pages:	2		

Title: Safety and Mission Assurance (S&MA) Plan

Description/Use: The MTN Safety and Mission Assurance (S&MA) Plan serves to identify and define the Contractor's plans and approach of accomplishing the Safety and Mission Assurance (S&MA) tasks to meet the specific MTN Program requirements in relation to the critical or key milestones during all phases of the Project life cycle. Include quality assurance aspects.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Baseline upon completion of Detailed Design Phase. Updates as required.

Remarks:

Contents:

The Safety and Mission Assurance (S&MA) Plan shall address S&MA philosophy, organization, approach and all the related processes/activities needed for program safety and mission success. The Contractor's SMAP shall include a traceability matrix of requirements to planned processes and verification for all S&MA requirements. This plan shall address the processes for reporting and tracking resolution of all corrective actions approved and resulting from investigations of mishaps, incidents, non-conformances, and anomalies. The plan shall document the key activities/tasks in each discipline supported by additional process details such as flow diagrams (where applicable), analysis tools/technologies, verification and validations system/tools to monitor the effectiveness and timely response of the product assurance system.

1. Safety and Mission Assurance Management: General information on how the S&MA function is managed, including integration of the S&MA function into the MTN design, development, risk management, test, ground, assembly, integration, and launch activities. The S&MA Plan shall include and address the following:
 - a. System Safety and Industrial Safety Plan: The Safety Plan shall contain sections related to System Safety, Ground Safety, and Industrial Safety. The Plan shall define the methodology and techniques for achieving the program safety requirements. This shall be a description of how the system safety and industrial/occupational and health safety programs shall be conducted to meet design specifications.

The System Safety Plan shall:

- i. Define the roles and responsibilities of personnel.

- ii. Define the required documentation, applicable requirements documents, and completion schedules for analyses and reviews. Address support for Reviews, Safety Working Group Meetings, and Technical Interchange Meetings (TIMs).
 - iii. Provide for early identification and control of hazards to personnel, facilities, support equipment, and the flight system during product development, including design, fabrication, test, transportation, and ground activities.
 - iv. Address compliance with industrial safety requirements imposed by NASA and OSHA design and operational needs and contractually imposed mission unique obligations. [Note: Safety and Health Plan may be delivered as an Appendix to the SMAP as a stand-alone document.
- b. Reliability and Maintainability Plan: The Reliability Program Plan shall explain how the contractor intends to implement design analyses into the final design. The explanation shall include the following:
 - i. The organizational responsibilities and functions for each task to be performed as part of the project.
 - ii. The point of contact for implementing and generating the necessary documents for design analysis.
 - iii. A list of subsystems and components for which trend analysis will be performed, including the parameters to be monitored.
 - iv. The execution and management plan for each task, including schedule.
 - v. Documentation, methods, scope, level of detail, and reporting specific to each task in the plan and for each element if there are differences.
 - vi. Alternate/additional methodologies (non-analysis, e.g., qual testing, modeling, inheritance assessment) for supporting analyses and addressing requirements.
- c. Materials and Processes Selection, Implementation, and Control Plan: For each Material and Process Requirement (MPR) in NASA-STD-6016C, Section 4, with the additions and prescribed changes below, the contractor shall prepare and implement an MPSCIP that states the MPR, identifies the degree of conformance under the subheading "Degree of Conformance," and identifies the method of implementation under the subheading "Method of Implementation."

The use of well-established contractor practices is encouraged to meet the intent of MPRs, and may be referenced by the MPSCIP, provided documentation of these practices is made available for review remotely by the Project M&P Engineer (MPE). Upon approval, the MPSCIP becomes the M&P implementation document used for verification (see NASA-STD-6016C, Section 4.1.1)

The MPSCIP shall include the contractor fastener management per contractor/industry standard policy (ref. NASA-STD-8739.14), including degree of conformance. The use of well-established contractor practices is encouraged, provided documentation of these practices are made available for review electronically for the Project MPE.

- d. Software Assurance Plan: The contractor shall document and implement a Software Assurance Plan and schedule compliant to contractor/industry standard (ref. NASA-STD-8739.8, NASA Software Assurance and Software Safety Standard). The plan will include the software assurance processes, procedures, tools and techniques to be used commensurate with the software classification and safety criticality assessment, with additional tailoring for each category of software (new, reused, Off-the-shelf, auto-generated code, etc..).

The Software Assurance Plan (SAP) shall address the following:

- a. Purpose and Scope
- b. Assurance Activities, Tools and Techniques by discipline:
 - a. Software Quality (process and product)
 - b. Software Safety
 - c. Software Reliability
 - d. Software Verification and Validation
 - e. Independent Verification and Validation (if applicable)
 - i. Software Assurance Program Metrics
 - ii. Problem Reporting and Corrective Action
 - iii. Assurance records, collection, maintenance, and retention
 - iv. Risk Management
 - a. Compliance Matrix
2. Quality Engineering/Assurance Plan The contractor shall document and implement their quality assurance plan including non-conformance process to track and disposition non-conformances, anomalies, and test failures.

Format: Contractor's native format.

End of Data Requirements Description DRD-SMA-02

5.38 Data Requirements Description DRD-SMA-03 (Safety Data Package (SDP))

Program:	MTN	DRD No.:	DRD-SMA-03
Data Type:	2	Date Revised:	Initial Release
Pages:	1		

Title: Safety Data Package (SDP)

Description/Use: The Safety Data Package (SDP) provides a description of the payload design to support hazard analysis results, hazard analysis method, and other applicable safety related information. The contractor shall include hazard analyses identifying the prelaunch, launch and flight hazards associated with the flight system, ground support equipment, and their interfaces. The contractor shall take measures to control or minimize hazards. In addition to identifying hazards, the SDP documents controls and verification methods for each hazard in Hazard Reports, which are included in a separate appendix. The analysis shall be updated as the hardware progresses through design, fabrication, and test. A list of hazardous/toxic materials with material safety data sheets and a description of the hazardous and safety critical operations associated with the payload shall be included in the final SDP. The safety assessment shall begin early in the program formulation process and continue throughout all phases of the mission lifecycle through safe separation from the launch vehicle. The spacecraft or instrument Project Manager shall demonstrate compliance with these requirements and shall certify to NASA Program/Project Office and the launch range, through the SDP, that all safety requirements have been met. Reference document for this DRD is *NASA-STD 8719.24 w/ Annex* (the NASA Payload Safety Requirements).

Initial Submission: Preliminary upon completion of Preliminary Design Phase**Submission Frequency:** Baseline upon completion of Detailed Design Phase. Final 60 working days prior to pre-ship review.**Remarks:****Contents:** The SDP content contains:

1. System description and background data necessary to the understand the system (includes interfaces, operations, processing and maintenance)
2. NASA-STD-8719.24, Volume 3, Attachment 1, Paragraph A1.2 provides a detailed description of the information required in the SDP.
3. The Final SDP shall also include appropriate forms as defined by the Payload Safety Working Group or governing safety review panel.

Format: Contractor's native format.**End of Data Requirements Description DRD-SMA-03**

5.39 Data Requirements Description DRD-SMA-04 (Failure Modes and Effects Analysis and Critical Items List / Single Point Failure)

Program:	MTN	DRD No.:	DRD-SMA-04
Data Type:	2	Date Revised:	Initial Release
Pages:	4		

Title: Failure Modes and Effects Analysis and Critical Items List / Single Point Failure

Description/Use: The purpose of the FMEA is to identify the possible failure modes associated with MTN design and their system effects and to drive hardware redesign (if needed), corrective actions or mitigations. The Critical Items List (CIL) indicates mission-critical hardware components (such as single-point failures) that cannot be designed out of the system and the actions that have been taken (or will be taken) to reduce the likelihood of MTN failure. In addition, the CIL cross-references to the Hazard Analysis Report. The second use of the FMEA is to assess design functions with respect to single failure tolerance and to indicate failure modes that may propagate across interfaces. The FMEA may also be used for troubleshooting.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Baseline upon completion of Detailed Design Phase. Final 20 working days prior to pre-ship review.

Remarks:

Contents:

FMEA Data Elements: Hardware item, function, mission phase, failure mode description, failure mode criticality, failure causes, failure effect on the item under analysis, failure effect at the sub-system level, failure effect on the MTN System, Time-to-worst case effect, Time to-detect, failure detection method, corrective action and/or mitigation. Time-to-detect and Time-to-worst case effect are expressed using the terminology below and are intended only as a best estimate.

1. Immediate: less than 1 second Seconds: 1 to 60 seconds
2. Minutes: 60 seconds to 60 minutes
3. Hours: 60 minutes to 24 hours
4. Days: 24 hours to mission completion

Failure mode criticality for the flight hardware and GSE FMEA is defined in the table below.

Category	Definition
1	Single point failure resulting in loss of life, permanently disabling injury, or vehicle.
1R	Redundant hardware elements, which if all failed, could cause permanently disabling injury loss of life or vehicle.
1S	Failure of emergency equipment and systems, hazard monitoring systems, or other hardware (e.g., burst disks, relief valves, communication devices) that could result in loss of or degraded capability to detect, combat, isolate, or operate when needed during a hazardous condition to preserve the life of the crew, prevent permanently disabling injury, or prevent the loss of the vehicle.
1SR	Redundant hardware elements, the failure of which renders inoperative a redundant system designed to monitor hazards or a system used to react to hazards; such hazards being sufficient to cause potential loss of life or vehicle.
2	Single failure that could result in a loss of mission or personnel injury or occupational illness requiring definitive/specialty hospital/medical treatment that results in loss of mission.
2R	Redundant hardware item that, if all failed, could cause a loss of mission or personnel injury or occupational illness requiring definitive/specialty hospital/medical treatment that results in loss of mission.
3	All others.

Critical Items List (CIL) Content and Definitions:

1. Items whose criticality category is 1, 1S, or 2.

2. Items whose criticality category is 1R, 1SR or 2R, AND that fail any of the Requirement Screens: A, B, or C.
3. Critical Items List Data Elements: Hardware Item Name, Identifier, Failure Mode, Cross reference to FMEA, Failure Mode Criticality Category, and its cross reference to the hazard analysis report for criticality 1, 1R, and 1S items.
4. For redundant items: Indication of Pass or Fail for Requirement Screens A, B, and C. A data entry column for Retention Rationale.

Requirement Screens:

Criticality 1R, 1SR and 2R failure modes are assessed for compliance with the requirement screens.

Screen A

The redundant hardware items (active and standby) are capable of checkout and verification during ground processing. This screen is not applicable to pyrotechnic devices, excluding electrical control circuitry.

Screen B

Loss of a redundant item is readily detectable during flight by the flight or ground crew (including automated detection). If time for corrective action exceeds time to effect, that failure mode shall be deemed not “readily detectable” and Screen B shall be shown as “FAIL.” (This screen is not applicable for standby items or items that are not turned on or are otherwise inactive until called upon for use.)

Screen C

Screen C shall indicate whether the loss of a critical function due to a failure of redundant hardware items can be the result of a single credible event or failure (e.g., contamination, fire, explosion, cable/wire failures,).

This screen shall be shown as “FAIL” when a single credible event or failure could result in the loss of a function. If a failure mode “FAILS” this screen, the common cause should be documented as part of the Screen C data field.

Definition of the Retention Rationale Prepared for the CIL:

A rationale for retaining a critical item in the hardware indicates those measures taken or planned in order to reduce the likelihood of failure.

These may include and are not limited to specific design features, design margin, operational workarounds, test results, planned inspections or tests, preventative and/or maintenance plans and flight history data.

FMEA Summary Report

1. Executive Summary – A one page summary of results
2. Table of Contents
3. Purpose and Scope of the analysis

4. Description of the System Analyzed
5. The Functional Block Diagram
6. Ground Rules and Assumptions
7. Summary of approach used. A brief summary of sections from handbooks or standards that were relevant to this analysis and were applied.
8. Subsystem Summary pages (for Flight Hardware and GSE)
9. A summary page for each subsystem shall contain:
 - a. A description of the function of the subsystem
 - b. Observations about the failure modes of the sub-system
 - c. A brief table of failure modes, effects, and causes for the Crit 1, 2 and 1S failure modes associated with the sub-system
10. Conclusions, Observations, and Recommendations

FMEA Worksheets

Sections: (1) Flight Hardware, (2) GSE

1. Worksheet files within sections broken down by sub-systems
2. Sub-System FMEA worksheets arranged by assemblies and containing line items for components
3. FMEA worksheets shall contain columns for the FMEA data elements.
4. FMEA line items on the worksheet shall be sortable into sections according to failure mode criticality (starting with criticality category 1 and in descending order of criticality down to criticality 3.)

CIL Worksheets

Sections: (1) Flight Hardware, (2) GSE

1. Worksheet sections broken down by sub-systems
2. Sub-System CIL worksheets arranged by assemblies and containing line items for components.
3. CIL worksheets shall contain columns for the Critical Item data elements.
4. CIL worksheets shall contain a column for retention rationale.
5. CIL line items on the worksheet are to be sortable according to failure mode criticality (starting with criticality category 1, followed by 1S, and 2).

Format: Contractor's native format.

End of Data Requirements Description DRD-SMA-04

5.40 Data Requirements Description DRD-SMA-05 (EEE Parts Management Plan)

Program:	MTN	DRD No.:	DRD-SMA-05
Data Type:	3	Date Revised:	Initial Release
Pages:	2		

Title: Electrical, Electronic, and Electromechanical (EEE) Parts Management Plan

Description/Use: The plan shall describe the implementation of the program to select and control EEE parts to support the required mission life and environment. The plan shall define and document the Contractor's requirements, system, and implementation plan for controlling the selection, acquisition, traceability, testing, handling, packaging, storage, and application of EEE parts for flight and critical ground support equipment.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Baseline upon completion of Detailed Design Phase. Final 20 working days prior to pre-ship review.

Remarks:

Contents:

NOTE: For this DRD, a Contractor's standard document that has been used successfully for similar space flight missions can be submitted as long as it meets the intent of the content requirements of this deliverable.

The plan shall address the following:

1. The Parts Control Board, its composition and responsibilities
2. Selection requirements, including parts grade or quality level, definitions for standard and non-standard parts, approval process for non-standard parts including non-standard part approval requests (NSPARs)
3. Qualification requirements for standard and non-standard parts
4. Requirements for de-rating of parts
5. Procurement requirements, including selection and monitoring of parts manufacturers, qualification and screening data, certificates of conformance to parts specifications, parts obsolescence management, and the approach that will be used to mitigate the risk of counterfeit parts
6. Radiation hardness requirements for TID, SEE, and DD
7. Review/approval process for custom devices
8. Review/approval requirements for the use of parts greater than five years old, including any re-screening requirements
9. Requirements for review and retention of qualification and screening data
10. Prohibited materials, such as tin platings/finishes with less than 3% lead, and mitigation methods when their use is unavoidable
11. Requirements for receiving, storage, handling, and traceability of EEE parts

12. The closed-loop GIDEP review process to prevent the use of unreliable and non-conforming parts

Format: Contractor's native format.

End of Data Requirements Description DRD-SMA-05

Released Version

5.41 Data Requirements Description DRD-SMA-06 (Parts Identification List)

Program:	MTN	DRD No.:	DRD-SMA-06
Data Type:	3	Date Revised:	Initial Release
Pages:	1		

Title: Parts Identification List

Description/Use: The EEE Parts Identification List (PIL) identifies the EEE parts that are proposed for use (As-Designed Parts List) or used (As-Built Parts List) in flight hardware and approved by the Parts Control Board.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Final as-built list 20 working days prior to pre-ship review.

Remarks:

Contents:

The EEE Parts Identification List shall contain the following information:

1. Flight component identity to the circuit board level
2. Complete part number (i.e., DLA military specification/drawing part number, or source control drawing part number, with all suffixes)
3. Manufacturer's Generic Part number
4. Manufacturer (not distributor)
5. Part Description (please include meaningful detail)
6. Federal Source Code or Federal Stock Class (FSC)
7. Procurement Specification (i.e., DLA military specification/drawing or source control drawing)
8. Comments and clarifications, as appropriate, including approval status of non-standard parts and waivers
9. Estimated quantity required (As-Designed Parts List, for procurement forecasting)
10. Reference designator (As-Built list only)
11. Lot date code/serial number (As-Built list only)

Format: Contractor's native format.

End of Data Requirements Description DRD-SMA-06

5.42 Data Requirements Description DRD-SMA-07 (Parts Stress/De-Rating Analysis)

Program:	MTN	DRD No.:	DRD-SMA-07
Data Type:	3	Date Revised:	Initial Release
Pages:	1		

Title: Parts Stress/De-Rating Analysis

Description/Use: The de-rating, when applied to certain EEE parts, improves the reliability of space flight systems by stressing the parts at a significant margin below their maximum ratings. The Contractor shall analyze the expected stresses for each EEE part that is proposed for use in flight hardware as compared to the maximum stress values derived from the de-rating criteria for the part.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Final upon completion of Detailed Design Phase

Remarks:

Contents:

The report shall contain the following:

1. Table of Contents, Purpose, Scope
2. Applicable Documents, including list of schematic diagrams
3. Tabulated Stress/De-Rating Analysis containing the following data:
 - a. Flight component identity to the circuit board level
 - b. Part type
 - c. Circuit reference designator
 - d. Specified maximum rating for each parameter to be de-rated for the part type
 - e. De-rating factor
 - f. De-rated parameter rating
 - g. Maximum stress
 - h. Margin between maximum stress and de-rated rating
 - i. Whether part meets or fails to meet de-rating criteria
4. Conclusions, including justification for use of parts not meeting de- rating criteria

Format: Contractor's native format.

End of Data Requirements Description DRD-SMA-07

5.43 Data Requirements Description DRD-SMA-08 (Radiation Tolerance Report)

Program:	MTN	DRD No.:	DRD-SMA-08
Data Type:	3	Date Revised:	Initial Release
Pages:	1		

Title: Radiation Tolerance Report

Description/Use: This report shall contain the expected radiation environment for the planned mission(s) and sets the minimum radiation parameters for total ionizing dose (TID), single event effects (SEE), and displacement damage (DD) that the EEE parts/components must be able to survive. The report shall list the EEE parts/components selected for the design with information to substantiate that the parts/components were selected to survive the expected radiation environment of the mission(s). For SEEs, single event latch-up analysis shall demonstrate recovery without damage at worst-case rated voltages and temperatures for susceptible parts, and single event burn-out and gate rupture analysis power transistors operated in off mode shall be evaluated at worst case application voltages. Also of specific interest is memory / EDAC.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Final upon completion of Detailed Design Phase

Remarks:

Contents:

The report shall contain the following:

1. Table of Contents, Purpose, Scope
2. Description of the mission radiation environment and tolerance requirements for TID, SEE, and DD
3. Ground rules and assumptions
4. Summary of technique used and tools selected
5. Parts list showing analysis results and deviations from requirements
6. Conclusion, observations, and recommendations, including description of mitigations for parts that do not meet radiation tolerance requirements

Format: Contractor's native format.

End of Data Requirements Description DRD-SMA-08

5.44 Data Requirements Description DRD-SMA-09 (Nonconformance Processing and Corrective Action Plan and Nonconformance Reports)

Program:	MTN	DRD No.:	DRD-SMA-09
Data Type:	2	Date Revised:	Initial Release
Pages:	3		

Title: Nonconformance Processing and Corrective Action Plan and Nonconformance Reports

Description/Use: To identify and define the Contractor's plans and approach for accomplishing the Nonconformance Processing and Corrective Action activities related to the work performed under the contract and to deliver individual Nonconformance Reports.

Initial Submission: Preliminary plan upon completion of Preliminary Design Phase.

Submission Frequency: Final plan upon completion of Detailed Design Phase. Reports as needed.

Remarks:

Contents:

As a minimum, the subparagraphs below shall be addressed in the Nonconformance Processing and Corrective Action Plan to describe the philosophy and approach for implementation of the planned nonconformance processing and corrective action activities, including details of responsibilities and controls. This may reference the Contractor's current Quality System Manual, Problem Reporting and Corrective Action (PRACA) system, procedures, and work instructions, such as by an outline, cross-reference matrix, etc. The Contractor shall provide access to their referenced first-tier procedures for review by NASA.

The Contractor, subcontractors, suppliers, and vendors with Material Review Board (MRB) authority shall adhere to these requirements. Subcontractors and vendors without MRB authority shall have an AS9100 compliant nonconformance and corrective action system and have their nonconformances, which require an MRB disposition, entered and approved in the system of the next higher tiered organization that has MRB authority.

Nonconformance Reporting

Nonconformances involving flight and flight-equivalent hardware/software/firmware (both contractor furnished and Government furnished) shall be reported and processed starting with those that occur or are detected starting at contract award.

NOTE 1: Flight-equivalent hardware/software is defined as a non-flight component built, inspected and tested to flight component specifications and built with manufacturing processes which are identical to those used for flight equipment. (e.g. Qualification hardware)

NOTE 2: The term problem is used interchangeably with nonconformance and is all-inclusive covering failure to conform to drawings or specifications, unsatisfactory

conditions, any anomaly such as an unexpected event, hardware or software malfunction or failure, departures from established procedures or practices, and processes.

NOTE 3: NASA's definition of "repair": Operations performed on a non-conforming article or material, which reduces but does not completely eliminate a non-conformance to place the article or material in a usable and acceptable condition; requires review and concurrence by an authorized MRB, additional written procedures and additional operations.

NOTE 4: NASA's definition of "rework": The procedure applied to software, components, articles and materials that will completely eliminate the non-conformance and result in complete conformance to requirements, drawings, specifications, procedures, or contract. Requires only normal operations or procedures to complete or fix the software, article or material in accordance with the applicable documents and does not require additional written procedures.

1. Reporting requirements shall continue through the hardware/software/firmware's life cycle until termination of the contract.
2. The Contractor shall make available all nonconformance reports and activities performed to resolve the nonconformance to NASA in electronic format on a monthly basis or by providing access to the Contractor PRACA database.
3. Prior to Government acceptance of hardware or turnover to NASA for operation, all nonconformance data shall be provided to NASA in electronic format.
4. Ground Support Equipment (GSE), Flight Operations, and Ground Operations facility problems, shall be provided to NASA, within 5 business days, if the problem caused or could have potentially caused adverse operational, safety, or quality assurance consequences on flight or flight-equivalent hardware/software or mission success.
5. Any functional failure or defect that could have led to a functional failure of Crit 1/1R, 1S/1SR, or Crit 2/2R hardware or software shall be provided to NASA within 5 business days.
6. Any nonconformance occurring after initial Government acceptance of a similar Configured Item that indicts or could affect the safety or flight-worthiness of previously accepted hardware or software shall be provided to NASA within 5 business days.

Material Review Boards

After entry into the Contractor's nonconformance reporting system, a preliminary review by the Contractor shall determine if the nonconformance should be dispositioned without an MRB, such as dispositioning as: scrap, rework, return to supplier, or standard repair procedure (SRP). As a minimum, nonconformances that do not meet any of these dispositions shall be referred to the Contractor's Material Review Board (MRB).

1. The Contractor's MRB shall be composed of:
 - a. one, at a minimum, Contractor representative whose primary responsibility is engineering;
 - b. one Contractor representative whose primary responsibility is product quality;
 - c. one NASA Quality Representative whose primary responsibility is quality engineering (only required for Government owned hardware or hardware that may be delivered to or operated by the Government) (with the concurrence of NASA, the Contractor MRB can be authorized to operate without NASA Quality representation); and

- d. other personnel as deemed appropriate by the Contractor.
2. The Contractor's MRB may approve hardware/software dispositions such as: repair, downgrade, or use-as-is (when analysis or other data supports the determination that form, fit, and function are not impacted).
3. For a disposition of Closed - Unexplained Anomaly (root cause(s) cannot be determined), documentation shall include the following:
 - a. Methods of recovery and/or operational workarounds.
 - b. Proximate cause (direct) or most probable cause.
 - c. Actions taken to minimize the impact of the nonconformance or to work around the nonconformance.
 - d. The effect to the crew, vehicle or mission should the problem reoccur.
 - e. How the crew or ground would become aware of the situation.
 - f. Any impacts to FMEA/CIL or Hazard Analyses documentation.
4. The Contractor's MRB may establish Standard Repair Procedures (SRPs). Once authorized by the MRB, SRPs dispositions may proceed without MRB approval.
5. Nonconformances meeting any of the following criteria shall be elevated to the NASA MRB for disposition:
 - a. Use-As-Is or Repair dispositions constituting a waiver or deviation to a requirement, or impacting physical or functional interfaces.
 - b. Changes to a previously-approved disposition
 - c. A Repair or Use-As-Is decision resulting in an either a known or potential hazard control violation, or an inability to completely comply with hazard control verifications, as a result of the defect or failure. This includes if the Repair or Use-As-Is decision affects a hazard of a 3x5 risk ranking or results in elevating a hazard to a 3x5 or greater (class I or IV, etc.) risk ranking.

Documentation for Resolution and Disposition

Documentation of nonconformance records, associated actions performed, resulting information developed in accomplishment of the analysis, disposition, and closure.

Format: Contractor's native format.

End of Data Requirements Description DRD-SMA-09

5.45 Data Requirements Description DRD-SMA-10 (Materials Identification and Usage List/Material Usage Agreement)

Program:	MTN	DRD No.:	DRD-SMA-10
Data Type:	3	Date Revised:	Initial Release
Pages:	1		

Title: Materials Identification and Usage List/Material Usage Agreement

Description/Use: To identify materials outgassing and other materials properties to ensure mission success.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Baseline upon completion of Detailed Design Phase. Final 20 working days prior to pre-ship review.

Remarks:

Contents:

The MIUL shall identify all unique materials and processes used in the construction of the deliverable hardware.

1. With approval from the NASA MPE, for NASA-STD-6016C, Section 4.1.5 or equivalent industry standard, the contractor may use NASA forms or the contractor's equivalent forms in lieu of the MAPTIS format.
2. The MIUL shall include Thermal Vacuum Stability data, when appropriate, per NASA-STD-6016C, Section 4.2.3.6 or equivalent industry standard.
3. The MIUL shall include the vacuum bake-out schedule for materials that fail outgassing requirements or provide a separate MUA.
4. The contractor may use the NASA outgassing database (URL <http://outgassing.nasa.gov>) in addition to MAPTIS (URL <http://maptis.nasa.gov>).
5. The MIUL shall document, when appropriate, compliance to MSFC-STD-3029A or equivalent industry standard.
6. The MIUL shall clearly identify all materials that do not meet the requirements of NASA-STD-6016C or equivalent industry standard and require an MUA.
7. The MIUL shall include soldering flux and solvents used for cleaning flight electronic assemblies, other than isopropyl alcohol or deionized water.

The MUA package shall include the technical information required to justify the application. MUAs for stress corrosion shall include a Stress Corrosion Cracking Evaluation Form per MSFC-STD-3029A (see NASA STD 6016C) or equivalent industry standard and a stress analysis.

Format: Contractor's native format.

End of Data Requirements Description DRD-SMA-10

5.46 Data Requirements Description DRD-SMA-11 (Planetary Protection Plan)

Program:	MTN	DRD No.:	DRD-SMA-11
Data Type:	2	Date Revised:	Initial Release
Pages:	1		

Title: Planetary Protection Plan

Description/Use: To establish planetary protection compliance and methods for planetary protection mitigation

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: - Final upon completion of Detailed Design Phase

Remarks: The contractor should work with the NASA MTN team to satisfy this DRD through streamlined compliance paths (e.g. trajectory analysis).

Contents:

The contractor shall provide data on planetary protection implementation, including, cleanroom environments and cleanroom certification, inventory of organic materials and approach to planetary protection implementation shall be provided to permit independent evaluation of planetary protection approach via the MIUL.

The following items should be included in the plan:

1. Cleanroom Properties: Implementation data associated with assembly, integration and test of hardware (cleanliness levels, methods used to prevent degradation of cleanliness conditions)
 - a. Locations of cleanrooms used for assembly, integration, testing and transport of hardware
 - b. Certification data for cleanrooms used for assembly, integration and transport of hardware consistent with an ISO 8 cleanroom.
 - c. Deviations from ISO 8 cleanroom practices.
2. Organic Inventory via the MIUL
 - a. Identification of MIUL status, maturity and work to go
 - b. Final MIUL provided at launch
3. Transport of Hardware
 - a. Identification of hardware, including levels of cleanliness and approaches to cleanliness consistent with planetary protection category

Implementation data associated with transport of hardware (cleanliness levels, methods used to prevent degradation of cleanliness conditions).

Format: Contractor's native format.

End of Data Requirements Description DRD-SMA-11

5.47 Data Requirements Description DRD-SMA-12(Safety Requirements Compliance Checklist (SRCC))

Program:	MTN	DRD No.:	DRD-SMA-12
Data Type:	2	Date Revised:	Initial Release
Pages:	1		

Title: Safety Requirements Compliance Checklist (SRCC)

Description/Use: The Safety Requirements Compliance Checklist (SRCC) is the document that provides information to the Government regarding compliance to NASA payload safety requirements in support of launch site processing. The checklist indicates for each requirement whether the proposed design is compliant, non-compliant but meets intent, non-compliant, or if the requirement is not applicable. An indication other than compliant will include rationale. Reference document for this DRD is *NASA-STD 8719.24 w/ Annex* (the NASA Payload Safety Requirements).

Initial Submission: Preliminary upon completion of Preliminary Design Phase.

Submission Frequency: Final upon completion of Detailed Design Phase

Remarks:

Contents:

The contractor shall prepare an SRCC of all design, test, analysis, and data submittal requirements. The SRCC content includes:

1. Criteria and requirement.
2. System
3. Indication of compliance, noncompliance, or not applicable
4. Rationale for indications other than compliant
5. Resolution
6. Reference
7. Copies of Range Safety and NASA approved non-compliances, including waivers and equivalent levels of safety certifications

Format: Contractor's native format

End of Data Requirements Description DRD-SMA-12

5.48 Data Requirements Description DRD-COM-01 (Communication System Analysis Report (DWE))

Program:	MTN	DRD No.:	DRD-COM-01
Data Type:	2	Date Revised:	Initial Release
Pages:	2		

Title: Communication System Analysis Report (DWE)

Description/Use: The Communication System Analysis Report documents the results of analyses performed in support of all phases of the communication system hardware and firmware/software development and use, and requirement verification.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Final upon completion of Detailed Design Phase

Remarks: This DRD is for the “Direct With Earth” (DWE) system.

Contents:

All analysis reports shall address, at a minimum, all topics listed below. The Contractor shall standardize an analysis report format.

1. Cover page with title, document number for traceability, relevant part and serial numbers (if applicable), analysis cycle, indication of purpose
2. Executive Summary
3. Objectives/Motivation for the analysis
4. Study Requirements and Constraints
5. Assumptions and ground rules
6. Initializing Data
7. Analysis tools, tests, and Models used
8. Modeling technique
9. Internal Verification and Validation of the results (if applicable)
10. Uncertainty of the results
11. Description of sensitivities
12. Detailed results
13. Key findings/recommendations

The Communication System Analysis Report shall include:

1. Brief description of each subsystem and overall function
2. Performance matrix stating each quantitative parameter requirement and assumption for each subsystem and analysis approach
3. Radio Frequency Link Margin - Document the link budget for each RF link
4. RF Coverage Analysis– Report of RF link availability considering vehicle trajectory, antenna patterns, structural blockage, planet/lunar blockage, and link margin.

5. Radiation patterns –Planned performance at Detailed Design Phase and test measurements when available
6. Pointing Analysis – Analysis of antenna pointing accuracy and stability considering all sources of pointing errors
7. RF Analysis – Analysis to show RF system compliance relating to RF operations and constraints with the launch site, vehicles, and facilities
8. RF Interference Analysis – Analysis or measured data showing that no other vehicle subsystems or components prevent the reception, transmission, or operation of RF communication subsystems or components. Similarly, provide analysis or measured data showing that no RF communication subsystem or components interfere with the proper operation of other vehicle subsystems or components, including simultaneous operation with other communication links.

Format: Contractor's native format. Link Budget submitted using NASA provided template per MTN-402-REQ-0001 (c).

End of Data Requirements Description DRD-COM-01

5.49 Data Requirements Description DRD-COM-02 (Communication System Analysis Report (UHF Proximity))

Program:	MTN	DRD No.:	DRD-COM-02
Data Type:	2	Date Revised:	Initial Release
Pages:	2		

Title: Communication System Analysis Report (UHF Proximity)

Description/Use: The Communication System Analysis Report documents the results of analyses performed in support of all phases of the communication system hardware and firmware/software development and use, and requirement verification.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Final upon completion of Detailed Design Phase

Remarks: This DRD is for the legacy UHF Proximity link system.

Contents:

All analysis reports shall address, at a minimum, all topics listed below. The Contractor shall standardize an analysis report format.

1. Cover page with title, document number for traceability, relevant part and serial numbers (if applicable), analysis cycle, indication of purpose
2. Executive Summary
3. Objectives/Motivation for the analysis
4. Study Requirements and Constraints
5. Assumptions and ground rules
6. Initializing Data
7. Analysis tools, tests, and Models used
8. Modeling technique
9. Internal Verification and Validation of the results (if applicable)
10. Uncertainty of the results
11. Description of sensitivities
12. Detailed results
13. Key findings/recommendations

The Communication System Analysis Report shall include:

1. Brief description of each subsystem and overall function
2. Performance matrix stating each quantitative parameter requirement and assumption for each subsystem and analysis approach
3. Radio Frequency Link Margin - Document the link budget for each RF link
4. RF Coverage Analysis– Report of RF link availability considering vehicle trajectory, antenna patterns, structural blockage, planet/lunar blockage, and link margin.

5. Radiation patterns –Planned performance at Detailed Design Phase and test measurements when available
6. Pointing Analysis – Analysis of antenna pointing accuracy and stability considering all sources of pointing errors
7. RF Analysis – Analysis to show RF system compliance relating to RF operations and constraints with the launch site, vehicles, and facilities
8. RF Interference Analysis – Analysis or measured data showing that no other vehicle subsystems or components prevent the reception, transmission, or operation of RF communication subsystems or components. Similarly, provide analysis or measured data showing that no RF communication subsystem or components interfere with the proper operation of other vehicle subsystems or components including simultaneous operation with other communication links.

Format: Contractor's native format. Link Budget submitted using NASA provided template per MTN-402-REQ-0001 (Mars Telecommunications Network (MTN) Objectives and Requirements).

End of Data Requirements Description DRD-COM-02

5.50 Data Requirements Description DRD-COM-03 (Communication System Analysis Report (Advanced Proximity))

Program:	MTN	DRD No.:	DRD-COM-03
Data Type:	2	Date Revised:	Initial Release
Pages:	2		

Title: Communication System Analysis Report (Advanced Proximity)

Description/Use: The Communication System Analysis Report documents the results of analyses performed in support of all phases of the communication system hardware and firmware/software development and use, and requirement verification.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Final upon completion of Detailed Design Phase

Remarks: This DRD is for the Advanced proximity link system.

Contents:

All analysis reports shall address, at a minimum, all topics listed below. The Contractor shall standardize an analysis report format.

1. Cover page with title, document number for traceability, relevant part and serial numbers (if applicable), analysis cycle, indication of purpose
2. Executive Summary
3. Objectives/Motivation for the analysis
4. Study Requirements and Constraints
5. Assumptions and ground rules
6. Initializing Data
7. Analysis tools, tests, and Models used
8. Modeling technique
9. Internal Verification and Validation of the results (if applicable)
10. Uncertainty of the results
11. Description of sensitivities
12. Detailed results
13. Key findings/recommendations

The Communication System Analysis Report shall include:

1. Brief description of each subsystem and overall function
2. Performance matrix stating each quantitative parameter requirement and assumption for each subsystem and analysis approach
3. Radio Frequency Link Margin - Document the link budget for each RF link
4. RF Coverage Analysis– Report of RF link availability considering vehicle trajectory, antenna patterns, structural blockage, planet/lunar blockage, and link margin.

5. Radiation patterns –Planned performance at Detailed Design Phase and test measurements when available
6. Pointing Analysis – Analysis of antenna pointing accuracy and stability considering all sources of pointing errors
7. RF Analysis – Analysis to show RF system compliance relating to RF operations and constraints with the launch site, vehicles, and facilities
8. RF Interference Analysis – Analysis or measured data showing that no other vehicle subsystems or components prevent the reception, transmission, or operation of RF communication subsystems or components. Similarly, provide analysis or measured data showing that no RF communication subsystem or components interfere with the proper operation of other vehicle subsystems or components including simultaneous operation with other communication links.

Format: Contractor's native format. Link Budget submitted using NASA provided template per MTN-402-REQ-0001 (Mars Telecommunications Network (MTN) Objectives and Requirements Document).

End of Data Requirements Description DRD-COM-03

5.51 Data Requirements Description DRD-COM-04 (Spectrum Management and Compliance Information)

Program:	MTN	DRD No.:	DRD-COM-04
Data Type:	2	Date Revised:	Initial Release
Pages:	1		

Title: Spectrum Management and Compliance Information

Description/Use: Spectrum management information is required to request frequency authorization for each communication link in accordance with NPD 2570.5E NASA RF Spectrum Management and National Telecommunications and Information Administration (NTIA) Manual of Regulations & Procedures for Federal Radio Frequency Management (May 2013 Edition, May 2014 Revisions) Chapter 10. NASA will use this information to request frequency authorization with the NTIA for all RF systems. Compliance with radio frequency allocation and conditions of assignment for frequency spectrum usage must be approved by the International Telecommunication Union (ITU) and NTIA.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Baseline upon completion of Detailed Design Phase. Final 20 working days prior to pre-ship review.

Remarks:

Contents:

The Spectrum Management information shall include the equipment characteristics, transmitter characteristics, receiver characteristics, and antenna characteristics as defined in Chapter 10 of the NTIA Manual of Regulations & Procedures for Federal Radio Frequency Management.

Format: Contractor's native format.

End of Data Requirements Description DRD-COM-04

5.52 Data Requirements Description DRD-FO-01 (Operations Data Book)

Program:	MTN	DRD No.:	DRD-FO-01
Data Type:	2	Date Revised:	Initial Release
Pages:	1		

Title: Operations Data Book

Description/Use: The Operations Data Book (ODB) is the single authoritative source of properly validated data which most accurately and completely describes the MTN operational performance capabilities and limitations. This document is used by the program, engineering, ground operations, flight operations, training, and contingency teams as a vehicle operating guide for ground operations planning, flight attitude planning, trajectory design, center of gravity management, consumables management, systems analysis, procedures development, flight rules development, flight techniques, training curriculum development, and stowage planning, understanding of fault management.

Initial Submission: Preliminary upon completion of Detailed Design Phase

Submission Frequency: Updated at major milestones as identified in DRD-PM-02 (Integrated Master Schedule). Final 20 working days prior to Acceptance Review.

Remarks:

Contents:

The ODB shall address the following content (separate volumes for each of the four major areas is recommended):

1. MTN Spacecraft Systems Performance and Constraints Data
 - a. Vehicle system configurations and coordinate systems
 - b. Constraints and operational limitations
 - c. Operational performance and operational data
2. MTN Spacecraft Systems Contingency Analysis Data
3. MTN Flight Capability Envelopes
 - a. Center of Mass Envelopes
 - b. Thermal envelopes
 - c. Translational Delta-V Capabilities
 - d. Attitude Control System Delta-V Capabilities
 - e. Sensed Acceleration Limits
4. MTN On-orbit Structural Envelopes

The content to be included in each volume shall be coordinated with the NASA subsystem managers to ensure the ODB contains the appropriate level of detailed information to meet the users' needs.

Format: Contractor's native format.

End of Data Requirements Description DRD-FO-01

Released Version

5.53 Data Requirements Description DRD-FO-02 (Flight Operations Procedures and Procedures Data)

Program:	MTN	DRD No.:	DRD-FO-02
Data Type:	2	Date Revised:	Initial Release
Pages:	2		

Title: Flight Operations Procedures and Procedures Data

Description/Use: The procedures and supporting procedures data will support mission design finalization, nominal and off-nominal flight operations implementation, and identify vehicle constraints, operational envelope violations, or deviations from verified capabilities. These Flight Operations Procedures and Procedures Data are used to refine the mission design and/or operations to ensure a viable mission implementation. Documentation of vehicle constraints, limitations and/or performance margins and assumptions will support development of Flight Operations products and mission execution.

Initial Submission: Preliminary upon completion of Detailed Design Phase

Submission Frequency: Updated at major milestones as identified in DRD-PM-02 (Integrated Master Schedule). Final 20 working days prior to Acceptance Review.

Remarks:

Contents:

1. The contractor shall update Flight Operations Procedures Data to implement the required operations of MTN. This documentation shall include the following:
 - a. Implementation and allocation of operations between automation, ground commanding, and flight operations
 - b. Trajectory design evaluation, assumptions and constraints for both nominal and abort conditions
 - c. Evaluation of integrated vehicle and subsystem performance drivers with respect to the mission design, including procedures for operator interaction with subsystems.
 - d. Vehicle automated sequencing and operator interaction procedure documentation
 - e. Integrated vehicle timeline products reflecting manual activity, automation, and integrated subsystem performance and procedures
2. The contractor shall provide subsystem and integrated vehicle procedures to facilitate appropriate operational implementations and flight specific solutions and provide documentation of assumptions, inputs, and results. Supporting procedures data documentation will be provided incorporating each of the following for integrated vehicle performance:
 - a. Assumptions, initial conditions, and performance parameters
 - b. Documentation of the case matrix including narrative discussion or rationale supporting the development of the procedures

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- c. Plots, data results, and/or numerical files for resulting procedures with identified intended use
 - d. Analysis of the deviations from procedure and conclusions to the viability of the mission implementation
 3. The contractor shall provide information for the development of operational techniques, strategies, and trades including abort coverage, refinement of mission constraints, and/or identification of mission assumptions necessary to successfully complete the mission objectives or implement sufficient responses to contingency situations.
 4. The contractor shall provide procedures with systems integration and integrated vehicle performance assessments accounted for to ensure the vehicle level operations reflect the subsystem capabilities, vehicle design limitations, and cross-system interactions, effects & constraints as necessary.

Format: Contractor's native format.

End of Data Requirements Description DRD-FO-02

5.54 Data Requirements Description DRD-FO-03 (Mission Operations Plans)

Program:	MTN	DRD No.:	DRD-FO-03
Data Type:	2	Date Revised:	Initial Release
Pages:	2		

Title: Mission Operations Plans

Description/Use: To provide definition of the contents of the Mission Operations Plan. It is a reference for the development of detailed activity plans both before launch and during the mission.

Initial Submission: Preliminary upon completion of Preliminary Design Phase

Submission Frequency: Updated at major milestones as identified in DRD-PM-02 (Integrated Master Schedule). Final 20 working days prior to Acceptance Review.

Remarks:

Contents:

The Mission Operations Plan shall include descriptions of the following:

1. Mission System Architecture. This will include the definition of operational scenarios and operational processes, core team roles and responsibilities, and interface definitions.
2. Mission Operations Concept. The Mission Operations Concept describes the baseline plan for conducting flight operations.
3. Baseline strategy for successfully achieving the mission objectives, as documented in DRD-MD-01, Mission Design Plan/Concept of Operation) within the capabilities and constraints of the flight system and mission system. Also describe the integration of the driving mission requirements and objectives, including a basic description of the mission components and how they are used to achieve those requirements.
4. Mission System Training Plan. The MS Training Plan describes how the Contractor will train personnel (both Contractor and NASA) to conduct mission operations and is compatible with the MS V&V Plan. This plan addresses the methodologies, constraints, responsibilities, and schedules associated with both Individual and Flight Team training as well as the methodologies for tracking Individual and Flight Team Certification. The plan addresses preparation for all phases of mission activity and is intended to be revised throughout the life of the mission with evolving training strategies as mission operations proceeds.
5. Ground Data System Integration, Test and Deployment (GDS ITD) Plan. The GDS ITD Plan describes both the integration approach and test program for system level verification of GDS capabilities and functionality, and the approach for deploying the verified GDS software to the proper project venues. The plan defines the test methodology utilized to verify all internal and external capabilities, interfaces (SISs), and products required by the MS, including where insight and oversight are used. The plan also includes the GDS "Test As You Fly" exceptions and risk assessment and

retirement/mitigation, approach to be used in the course of verification activities. The GDS ITD plan describes the process and participants involved in deploying GDS software to project venues. The plan also includes the approach for customer adaptation for operational needs. The plan describes the roles and responsibilities of test and deployment personnel as well as interfaces among project and external (multi-mission) entities. The GDS ITD plan covers all GDS deliveries.

6. Ground System Compatibility Test Plan. Demonstrate that the Mission System developed by the Contractor is able to interface with the Flight System in an operational way (e.g., able to radiate commands that are accepted and acted upon correctly by the spacecraft, able to receive telemetry at all possible telemetry configurations and rates and decipher them correctly, etc.)
7. Deep Space Network (DSN) Service Agreement to document the agreement between the MTN Contractor and the DSN for DSN services.

Format: Contractor's native format.

End of Data Requirements Description DRD-FO-03

Appendix A Abbreviations and Acronyms

ADP	Acceptance Data Package
AI&T	Assembly, Integration, & Test
APID	Application Program Interface Division/Definition
ATP	Authority to Proceed
CAD	Computer Aided Design
CDM	Configuration and Data Management
CDMP	Configuration and Data Management Plan
CEI	Component End Item
CI	Configuration Item
CIL	Critical Items List
CM	Configuration Management
CO	Contracting Officer
ConOps	Concept of Operations
COTS	Commercial-off-the-shelf
CPU	Central Processing Unit
CSCI	Computer Software Configuration Item
CWBS	Contractor Work Breakdown Structure
DD	Displacement Damage
DDB	Design Data Book
DLA	Destructive Lot Acceptance
DM	Data Management
DPD	Data Procurement Document
DRD	Data Requirements Description
DSN	Deep Space Network
DWE	Direct with Earth
EDAC	Error Detection and Correction
EDL	Entry, Descent, & Landing
EEE	Electric, Electronic, & Electromechanical
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EOL	End of Life
EOM	End of Mission
EOR	Earth Orbit Raise
FMEA	Failure Modes and Effects Analysis
FSC	Federal Stock Class
GD&T	Geometric Dimensioning and Tolerancing
GDS	Ground Data System

GIDEP	Government-Industry Data Exchange Program
GQA	Government Quality Assurance
GSE	Ground Support Equipment
ICD	Interface Control Document/Drawing
IMS	Integrated Master Schedule
IP	Intellectual Property
IT	Information Technology
ITD	Integration, Test and Deployment
MAPTIS	Materials And Processes Technical Information System
MDD	Mission Design Document
MEL	Master Equipment List
MIUL	Materials Identification and Usage List
MPE	Materials and Processes Engineer
MPR	Material and Process Requirement
MPSCIP	Materials and Processes Selection, Control, and Implementation Plan
MRB	Material Review Board
MTN	Mars Telecommunications Network
MUA	Material Usage Agreement
NASA	National Aeronautics and Space Administration
NIST	National Institute of Standards & Technology
NSPAR	Non-Standard Part Approval Request
PEL	Power Equipment List
PIL	Parts Identification List
PMP	Project Management Plan
PRACA	Problem Reporting and Corrective Action
PWS	Performance Work Statement
RCS	Reaction Control System
RMP	Risk Management Plan
S&MA	Safety and Mission Assurance
SAP	Software Assurance Plan
SDP	Safety Data Package
SEE	Single Event Effects
SRCC	Safety Requirements Compliance Checklist
SRP	Standard Repair Procedure
SRS	Software Requirements Specification
TID	Total Ionizing Dose
TIM	Technical Interchange Meeting
TPPs	Technical Performance Parameters
UHF	Ultra-High Frequency

V&V	Verification & Validation
VCN	Verification Closure Notice
WBS	Work Breakdown Structure
WCA	Worst Case Analysis

Released Version